

Pollution-Prevention Actions and Next-Year Toxic Release Reduction in U.S.

Manufacturing

Final Report

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Code and Data Availability

GitHub repository (code, notebooks, data dictionary, and sample files):

<https://github.com/somndas3/qm640-tri-pollution-prevention-capstone>

Final analysis notebook: notebooks/02_QM640_TRI_2020_2024_final.ipynb

Exploratory data analysis notebook:

notebooks/02_EDA_QM640_Synopsis_TRI_2020_2024.ipynb

Data preparation pipeline: src/data_prep.py

Raw and processed data, view-only (Google Drive):

https://drive.google.com/drive/folders/1oPNHYCAvGwc7uQ1iIP2a_VlZdLnFJuid?usp=sharing

Original data source (U.S. Environmental Protection Agency): <https://www.epa.gov/toxics-release-inventory-tri-program/tri-basic-plus-data-files-calendar-years-1987-present>

The ten raw source files total 535.6 MB, which exceeds practical repository limits, so they are shared through Google Drive rather than GitHub. [Appendix B](#) lists the size and SHA-256 checksum of every file, and [Figures D1](#) to [D4](#) in [Appendix D](#) show the shared folder structure and processed files. The final notebook ran from top to bottom without errors. Random elements were fixed with a seed of 42 so that the tuned model and bootstrap intervals can be reproduced.

Abstract

U.S. manufacturing facilities report pollution-prevention actions to the Environmental Protection Agency (EPA) every year, but environmental health and safety (EHS) teams lack an evidence-based way to decide which filings deserve review. This study examined whether reported actions are associated with meaningful next-year toxic-release reduction and whether action-year information can rank records by risk of not reducing. Data were Toxics Release Inventory (TRI) Basic Plus Form R records for reporting years 2020 through 2024, analyzed at the facility-chemical-year level; meaningful reduction meant a next-year on-site release of no more than 90% of the current-year release. Chi-square tests, facility-clustered logistic regression, and Holm-corrected comparisons addressed explanation; an elastic-net logistic regression tuned on 123,535 development records was evaluated once against a baseline on a sealed 2023-to-2024 holdout. Reduction rates rose from 27.5% to 35.9% across transitions and varied by 15 percentage points across subsectors, but effect sizes were small. After adjustment, reporting any action was associated with 14% higher odds of reduction (odds ratio [OR] = 1.14, 95% confidence interval [CI] [1.06, 1.23], $p < .001$). Only material modification and operating practice or training exceeded the no-action rate after correction. On the holdout, the model reached an area under the receiver operating characteristic curve (AUC) of .523, 95% CI [.515, .532], versus .502 for the baseline, and 73.9% of the top 5% of ranked records did not reduce against a 68% base rate. The findings support review triage and prioritized evaluation of the strongest action types as decision support, not automation or enforcement.

Keywords: Toxics Release Inventory, pollution prevention, source reduction, logistic regression, elastic net, out-of-time validation, environmental health and safety

Pollution-Prevention Actions and Next-Year Toxic Release Reduction in U.S.

Manufacturing

Every year, thousands of U.S. manufacturing plants report their toxic releases and the steps they took to release less. An environmental health and safety (EHS) team can review only a small share of these filings. This study tested whether reported pollution-prevention actions carry real information about next-year release reductions and whether that information can help such a team decide where to look first. It is an association and screening study, not a causal one.

Background and Context

The U.S. Environmental Protection Agency (EPA) collects these filings through the Toxics Release Inventory (TRI), a public program whose data support open study of industrial environmental behavior (U.S. Environmental Protection Agency [EPA], 2025b, 2025c, 2025d). The Pollution Prevention Act of 1990 favors preventing waste at the source over recycling, treatment, or disposal (EPA, 2025a), and facilities report such source-reduction actions with standard activity codes.

Plant managers, EHS leaders, sustainability leaders, and regulators need to know whether reported actions line up with lower releases the following year, and which kinds of action do so most strongly. A simple before-and-after comparison cannot answer this, because releases also move with production, reporting changes, and ordinary year-to-year variation.

Earlier TRI research studied disclosure, voluntary programs, and the long-run effects and profitability of prevention (Cagno et al., 2005; Gamper-Rabindran, 2006; Hamilton, 1995; King & Lenox, 2002; Konar & Cohen, 1997; Ranson et al., 2015). None of it gives an EHS team a ready-to-use way to screen this year's filings, which is the gap this project addresses.

Problem Statement

EHS and plant leaders lack a recent, transparent way to judge whether records carrying reported prevention actions are more likely to show a meaningful next-year reduction in the same chemical's on-site release. The objective of this study was to explain and predict $Y = \text{REDUCTION}_{10}$, equal to 1 when the next-year on-site release is no more than 90% of the current-year release, for U.S. manufacturing facility-chemical-year records, using $X =$ reported pollution-prevention actions, current release, production or activity ratio, manufacturing subsector, chemical hazard classification, and action year, over TRI reporting years 2020 through 2024. Success was evaluated with adjusted odds ratios (ORs) and facility-clustered 95% confidence intervals (CIs), and with holdout area under the receiver operating characteristic curve (AUC), Brier score, and top-ranked non-reduction rates against a simple baseline.

Purpose of the Study

The study aimed to explain which factors are associated with next-year reduction and to predict which records are unlikely to reduce, so that EHS teams can prioritize review without replacing engineering judgment. The overarching question was: *To what extent can recent EPA TRI information on reported pollution-prevention actions help explain and predict whether a facility will reduce the same chemical's on-site release by at least 10% in the following year?*

Research Questions

Four research questions (RQs) addressed context (RQ1), explanation (RQ2 and RQ3), and prediction (RQ4).

RQ1. Do meaningful next-year reduction rates differ across action-year transitions and manufacturing subsectors?

RQ2. After controlling for current release level, production or activity ratio, manufacturing subsector, chemical hazard classification, and action year, is reporting at least one pollution-prevention action associated with achieving a meaningful next-year reduction?

RQ3. Do meaningful next-year reduction rates differ between each reported pollution-prevention action type and records with no reported action?

RQ4. Can an interpretable model using only action-year information predict which facility-chemical records will fail to achieve a meaningful next-year reduction better than chance and better than a simple baseline model?

Contributions and Expected Value

Practical Contribution. A ranked review list and evidence on which reported action types show the clearest association with reduction, both built from public data that EHS teams already hold.

Technical Contribution. A design that separates adjusted explanation from out-of-time prediction, with facility-clustered inference and a sealed holdout evaluated once.

Value to Stakeholders. EHS teams gain a way to order a fixed workload, sustainability leaders gain peer-aware benchmarks, and regulators gain a candid view of what checkbox-style TRI data can support.

Literature Review

Literature Review Approach

Subject-matter sources on TRI disclosure, voluntary programs, and source reduction were found through Google Scholar and EPA documentation and were kept when they reported U.S. facility-level or firm-level evidence. Method sources were kept when they governed a specific design decision in this study.

Summary of Key Literature

[Table 1](#) summarizes the 10 key sources, and [Appendix G, Table G1](#) summarizes the supporting method and documentation sources. Author names in both tables link to the References.

Table 1

Literature Relevance Matrix for Key Sources

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	How It Supports the RQs or Project Decisions
Hamilton (1995)	TRI disclosure and markets	U.S. public firms named in the first TRI release	Event study	Firms reporting TRI releases had negative abnormal stock returns.	TRI reporting carries real consequences, so reported behavior is worth studying
Konar & Cohen (1997)	Community right-to-know	U.S. firms reporting to TRI	Panel analysis	Firms with the largest stock-price declines later cut emissions more.	Links reported behavior with later releases
Gamper-Rabindran (2006)	EPA 33/50 voluntary program	Manufacturing plants in 48 states	Evaluation controlling for self-selection	No reduction in health-indexed emissions in several key industries; some apparent cuts were transfers to recyclers.	Raw reductions can mislead; supports adjustment, subsector controls, and no causal claims (RQ1, RQ2)
King & Lenox (2002)	Prevention versus treatment	U.S. manufacturing firms, TRI data	Panel regression	Profitable pollution reduction came from preventing waste, not treatment.	Focus on source-reduction codes
Cagno et al. (2005)	Cleaner production economics	134 prevention project reports	Comparative analysis	Project profitability differed across intervention types.	Compare the five action types separately (RQ3)
Ranson et al. (2015)	Pollution prevention outcomes	U.S. manufacturing facilities in TRI	Facility-level regression	Source-reduction activity was associated with lower releases.	Closest prior study; motivates RQ2 and RQ3 on recent data
Cameron & Miller (2015)	Clustered inference	Econometric practice	Practitioner guide	Ignoring within-cluster correlation understates standard errors.	Facility-clustered errors in RQ1 to RQ3
Holm (1979)	Multiple comparisons	Statistical theory	Step-down procedure	Controls family-wise error with more power than Bonferroni.	Correction for the five RQ3 comparisons
Zou & Hastie (2005)	Regularized regression	Statistical methodology	Elastic-net penalty	Combined lasso and ridge penalties handle correlated predictors.	Choice of the RQ4 elastic-net model
Steyerberg et al. (2010)	Prediction model evaluation	Methodological framework	Performance measures	Discrimination alone is not enough; calibration and usefulness matter.	RQ4 metrics: AUC, Brier score, calibration, lift

Note. The remaining sources (interval estimation, large-sample inference, regression to the mean, functional form, sample size, calibration, and EPA documentation) appear in [Appendix G, Table G1](#). AUC = area under the receiver operating characteristic curve; EPA = U.S. Environmental Protection Agency; RQ = research question; TRI = Toxics Release Inventory.

Public reporting changes behavior, but raw numbers can mislead. The first TRI disclosure moved share prices (Hamilton, 1995), and the firms punished hardest later cut emissions most (Konar & Cohen, 1997). Yet in the EPA's voluntary 33/50 program, some apparent reductions reflected transfers to recyclers rather than true elimination (Gamper-Rabindran, 2006). Prevention also varies by type. Profitable reduction came from preventing waste rather than treating it (King & Lenox, 2002), project profitability differed across interventions (Cagno et al., 2005), and source reduction was associated with lower releases (Ranson et al., 2015). These studies justify a narrow outcome, adjusted comparisons, and separate analysis of action types.

The methods literature shaped inference and prediction. Clustered standard errors prevent understated uncertainty when facilities repeat (Cameron & Miller, 2015), Holm's (1979) correction controls error across multiple comparisons, and with very large samples effect sizes matter more than p values (Lin et al., 2013). The elastic net suits correlated predictors (Zou & Hastie, 2005), and prediction models should be judged on calibration and decision usefulness as well as discrimination (Steyerberg et al., 2010; Van Calster et al., 2019).

Gap Addressed by This Study

Prior work shows that disclosure matters, that raw trends mislead, and that prevention types differ. It does not offer a recent, record-level screening tool or a candid estimate of how much predictive signal TRI action fields contain. This study addresses both.

Materials and Method

The study was a retrospective observational analysis of public EPA data with two paths: an explanatory path for RQ1 to RQ3 and a predictive path for RQ4. The final transition was sealed as an out-of-time holdout for the predictive evaluation.

Data Source

Data came from the EPA TRI Basic Plus Data Files (EPA, 2025c, 2025d): File 1A and File 2A for each reporting year from 2020 through 2024. File 1A supplies identifiers, the North American Industry Classification System (NAICS) code, unit, hazard flags, and on-site releases. File 2A supplies the production or activity ratio and up to four source-reduction codes. Files were joined within each year on the document control number ([Appendix B](#)), and the data dictionary appears in [Appendix E, Table E1](#).

Population and Unit of Analysis

The population was U.S. TRI-reporting manufacturing facilities with a primary NAICS code beginning with 31, 32, or 33, using Form R records only. The unit of analysis was the facility-chemical-year, forming four transitions from 2020 to 2021 through 2023 to 2024.

Inclusion and Exclusion Criteria

Records were included when they had a matching facility-chemical record in the following year, a current-year release above zero, and the same unit in both years. These rules reduced 315,581 raw rows to 163,501 eligible records ([Appendix E, Table E3](#)). Records missing in the next year were excluded rather than scored as reductions.

The development set held 123,535 records from 14,851 facilities, including 122,017 pound-reported records used for RQ2 and RQ4 and 1,518 gram-reported dioxin records analyzed separately. The 2023-to-2024 holdout contained 39,966 eligible records, of which 39,500 pound-reported records formed the RQ4 modeling holdout.

Outcome Definition

REDUCTION_10 equals 1 when $\text{release}_{t+1} \leq 0.90 \times \text{release}_t$ and 0 otherwise, and it is defined only when current release exceeds zero. The 10% threshold was fixed before any

modeling as a materiality cutoff, so it could not be tuned to improve results. A round 10% change is easy for managers to interpret and keeps small year-to-year movements from counting as reductions. Alternative thresholds were not tested. The rate was 31.48% in development and 32.05% in the holdout. For RQ4 the positive class was non-reduction, the outcome an EHS team would act on.

Predictors

Predictors were a reported-action indicator and five action-type indicators (material, product, process or equipment, inventory or material management, and operating practice or training), current release, the production ratio, the three-digit subsector, EPA hazard flags for carcinogen, persistent, bioaccumulative, and toxic (PBT), and metal status, and the transition. The 15 subsectors with at least 500 records were kept individually, and the rest formed Other Manufacturing. An action indicator of 0 means that no code was observed, not that no action occurred.

Data Preparation

Every cleaning filter was logged and reconciled against raw counts ([Appendix E, Table E2](#)). Duplicate filings were consolidated with a deterministic rule ([Appendix C](#)), unit labels were normalized, and 70 activity codes spanning an EPA coding change were mapped to the five action types. The holdout file was written without outcome columns.

Exploratory Data Analysis

Exploratory analysis used development data only. [Table 2](#) links each important finding to its interpretation and modeling consequence, and [Appendix A](#) gives the full discussion and figures.

Table 2*Exploratory Findings, Interpretation, and Modeling Consequences*

Finding	Interpretation	Modeling or Business Consequence
31.48% of development records met the reduction threshold; only 4.87% reported any action (Figure A1)	The outcome is not severely imbalanced, but the exposure is rare	No resampling; rare exposure, not class imbalance, is the main statistical constraint
Process or equipment modification rose from 1.32% to 2.13%, and material modification from 0.41% to 0.87%, exactly at the 2020-to-2021 code-scheme change (Figures A2, A3)	Part of the rise may be a reporting artifact rather than behavior	Codes mapped to five types; key association repeated without the 2020-to-2021 transition
Reduction rates rose from 27.54% to 35.86% across transitions and spanned 15.4 percentage points across subsectors	Time and industry both shift the outcome	Year and subsector used as RQ2 controls; formal tests reported under RQ1
Subsectors reporting the most actions were not those reducing most often (Figure A4)	An unadjusted action comparison is partly a comparison of industries	Adjusted model required; below-threshold subsectors grouped as Other Manufacturing
Raw gap of 3.61 points for any action; 7 of 19 covariates imbalanced beyond a standardized difference of 0.10 (Figures A5 to A7)	The raw gap is confounded	RQ2 answered only with an adjusted model
Current release: mean 24,737 lb, median 217 lb, skewness 34.6; nonmonotonic rates by release decile (Figures A8 to A10)	Raw scale is unusable; starting level shapes room to fall (regression to the mean)	Log transformation; current release forced into every model
U-shaped reduction rate across production-ratio deciles; 1.62% missing; extreme values (Figures 1, A11)	A linear term would miss the shape; a few extreme values could dominate	Centered quadratic term, median imputation with a missingness flag, 99th-percentile cap
122,017 pound-frame records from 14,836 facilities; 92.8% from multi-record facilities (Figure A3)	Records are not independent	Facility-clustered errors, facility bootstrap, and time-grouped validation
Dioxins reduce at 24.84% versus 31.56%; carcinogen status shows a 0.07-point difference (Figure A12)	Dioxins differ and use grams	Dioxins analyzed separately; hazard flags kept as controls
Cross-year attrition of 7.6% to 9.5%; higher among action-reporting records (Figure A13)	Selection into the matched sample is possible	Reported as a limitation; unmatched records never scored as reductions

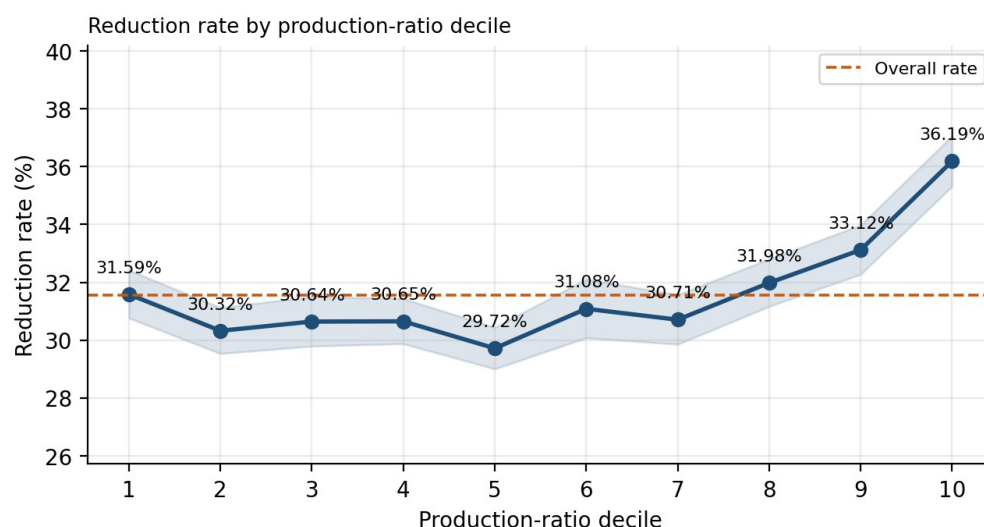
Note. Rates and counts refer to the development set. Figure 1 appears in the main text; Figures

A1 to A14 appear in Appendix A, where Figure A14 summarizes the 15 data-quality checks.

One finding changed the specification directly. The reduction rate was U-shaped across production-ratio deciles, lowest in the middle decile (29.72%) and highest in the top decile (36.19%), as Figure 1 shows. A centered quadratic term was therefore added.

Figure 1

Reduction Rate by Production-Ratio Decile in the Development Set



Note. Shaded bands are 95% Wilson confidence intervals; the dashed line is the overall development reduction rate.

Feature Engineering

Table 3 lists the final features and their use. RQ2 used the full adjustment set. The RQ4 model used 13 action-year predictors and excluded action year, because the holdout year was unseen in training. The locked RQ4 specification was limited to release, production, action, and chemical hazard features, so subsector indicators were not included and their added predictive value was not tested.

Table 3

Final Features, Construction, and Use by Research Question

Feature	Construction	Reason	RQ2	RQ4
Log current release	$\log(1 + \text{current-year release, lb})$	Skewness of 34.6; controls regression to the mean	Yes	Yes
Production ratio, centered	Missing values set to the development median (1.00); capped at the 99th percentile (3.15); centered at the pound-frame mean (1.0444)	Limits extreme values; centering reduces collinearity with the squared term	Yes	Yes

Feature	Construction	Reason	RQ2	RQ4
Production ratio squared	Square of the centered ratio	Captures the U-shape; supported by AIC	Yes	Yes
Production ratio missing	1 when the ratio was originally absent	Keeps information about missingness after imputation	Yes	Yes
Any reported action	1 when at least one valid source-reduction code is present	Primary RQ2 exposure	Yes	Yes
Five action-type indicators	One indicator per mapped action type; not mutually exclusive	Types may differ; pooling could hide that (RQ3)	No	Yes
Carcinogen and PBT flags	EPA classification flags	Hazard controls	Yes	Yes
Metal flag	EPA metal or metal-compound flag	Chemical-class signal	No	Yes
Manufacturing subsector	15 qualifying subsectors plus Other Manufacturing; reference NAICS 325 (chemical)	Industry confounding	Yes	No
Action-year transition	Reference 2020 to 2021	Time trend confounding; holdout year unseen in training	Yes	No
Identifiers	Facility, chemical, CAS number, chemical name	Traceability only	No	No

Note. In RQ4, the three continuous features were standardized inside each cross-validation training fold, and binary indicators were passed through unchanged. Gram-reported dioxin records were outside the pound-reported frame, so a dioxin flag was not modeled. AIC = Akaike information criterion; CAS = Chemical Abstracts Service; PBT = persistent, bioaccumulative, and toxic.

Leakage controls included an outcome-free holdout file, labels rebuilt only after locking, development-only preprocessing values, standardization within training folds, and folds grouped by transition. The imputation, cap, and centering constants were computed once on all development data rather than within each fold, a minor departure from a fully nested design.

Sample Size and Power Justification

Minimum requirements were 1,068 records for RQ1, 10,756 for RQ2, about 290 per action type for RQ3, and 1,269 for RQ4 ([Appendix E, Table E4](#)). All were met, and the locked RQ4 model had about 2,963 events per parameter. Because RQ3 was powered for 10-point differences, a nonsignificant action type does not show that the action has no association.

Research Questions and Hypotheses

Table 4 states the hypotheses, method, and decision rule for each question. Tests used $\alpha = .05$, RQ3 used Holm's (1979) correction, and RQ4 decisions used 95% facility-bootstrap intervals.

Table 4

Research Questions, Hypotheses, Methods, and Decision Rules

RQ	Null Hypothesis (H_0)	Alternative Hypothesis (H_a)	Method	Decision Rule
RQ1a	Meaningful reduction is independent of action-year transition	Reduction rates differ across transitions	Pearson chi-square test, Cramér's V, Wilson CIs	Reject H_0 if $p < .05$; judge size by V
RQ1b	Meaningful reduction is independent of manufacturing subsector	Reduction rates differ across subsectors	Pearson chi-square test, Cramér's V, Wilson CIs	Reject H_0 if $p < .05$; judge size by V
RQ2	The adjusted OR for reporting any action equals 1	The adjusted OR differs from 1	Logistic regression with facility-clustered standard errors	Reject H_0 if $p < .05$ and the 95% CI excludes 1
RQ3	For each action type, the reduction rate equals the no-action rate	The action-type rate differs from the no-action rate	Five 2×2 chi-square tests, rate differences, RRs, Holm correction	Reject H_0 for a type if Holm-adjusted $p < .05$
RQ4a	Holdout AUC of the elastic-net model equals .50 (chance)	Holdout AUC exceeds .50	Facility bootstrap CI for AUC on the sealed holdout	Reject H_0 if the 95% CI lower bound exceeds .50
RQ4b	Elastic-net holdout AUC does not exceed baseline AUC	Elastic-net holdout AUC exceeds baseline AUC	Paired facility bootstrap of the AUC difference	Reject H_0 if the 95% CI of the difference excludes 0

Note. AUC = area under the receiver operating characteristic curve; CI = confidence interval; OR = odds ratio; RR = relative risk.

Statistical Methods

RQ1 used Pearson chi-square tests with Cramér's V and Wilson intervals, because significance alone is uninformative with more than 120,000 records (Lin et al., 2013). RQ2 used logistic regression, whose coefficients convert into odds ratios, with standard errors clustered by facility because facilities contribute repeated records (Cameron & Miller, 2015). Diagnostics included variance inflation factors (VIFs), the condition number, and an Akaike information criterion (AIC) comparison of production-ratio forms. RQ3 compared each action type with the

no-action group in 2×2 chi-square tests, reporting rate differences, relative risks (RRs), and Holm-adjusted p values.

Model Development

The predictive path compared an unpenalized baseline logistic regression, using only the any-action indicator and log current release, with an elastic-net logistic regression using the 13 predictors in [Table 3](#) (Zou & Hastie, 2005). The elastic net stays interpretable, handles correlated action indicators, and resists overfitting. A grid search selected $C = 0.01$ with an L1 ratio of .1, a heavily penalized fit. Both models were refitted on all development data and locked. Flexible learners were not adopted, because reviewers need to know why a record is flagged.

Validation Strategy

The split was temporal because real use is forward-looking, and a random row-level split could place observations from the same facilities in both training and validation sets, producing an overly optimistic estimate of future performance. Tuning used leave-one-transition-out cross-validation within development data. Holdout labels were rebuilt from raw 2024 filings only after the design was locked, matching all 39,500 pound-reported records, and the holdout was used once. Holdout uncertainty came from 500 facility-level bootstrap resamples.

Evaluation Metrics

Explanatory results were judged by effect sizes and intervals. Predictive models were judged on AUC, Brier score, calibration, and screening lift among top-ranked records (Steyerberg et al., 2010; Van Calster et al., 2019). Accuracy was not a headline metric, because always predicting non-reduction is already correct for about 68% of records ([Appendix E, Table E5](#)).

Sensitivity Analyses

Six sensitivity checks tested the time period, the imputation choice, the functional form, facility clustering for RQ1 and RQ3, and separate handling of dioxins ([Appendix F, Table F1](#)).

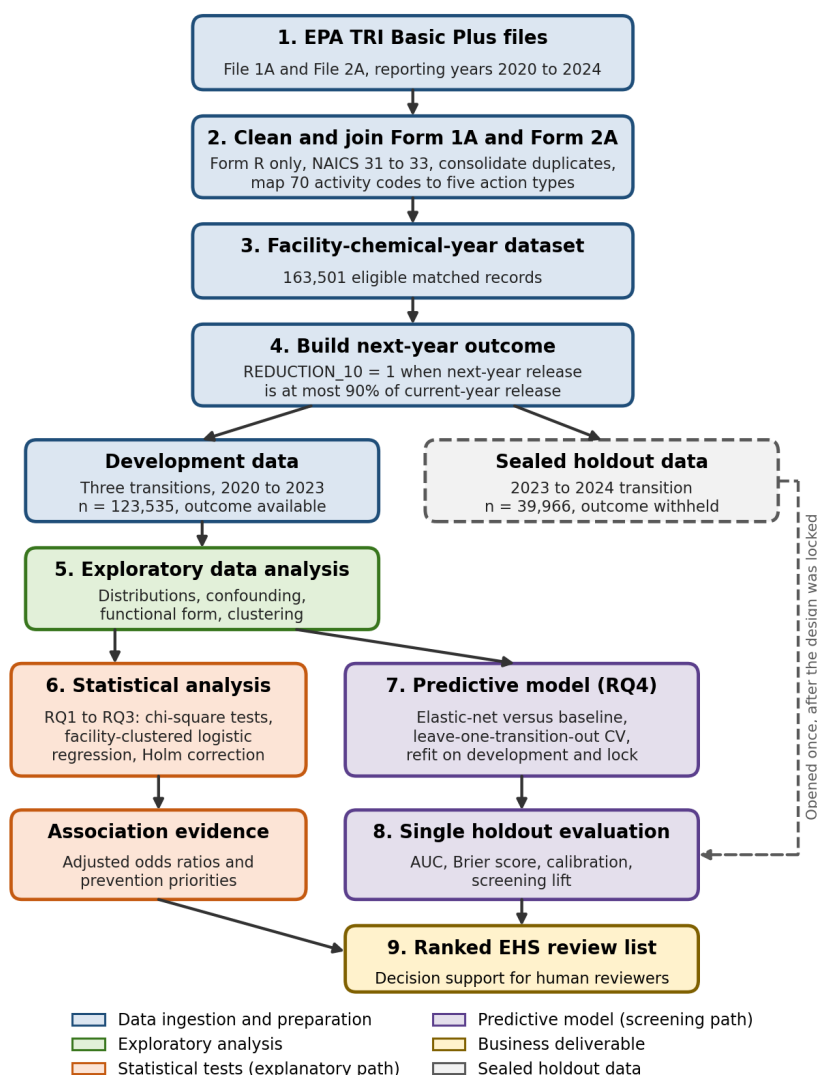
Architecture Diagram and Workflow

System Overview

The pipeline turns raw EPA files into two outputs: statistical evidence for RQ1 to RQ3 and a ranked review list from a locked screening model for RQ4. All stages ran in Python notebooks in Google Colab, with data on Google Drive and code on GitHub.

Architecture Diagram

As shown in [Figure 2](#), the cleaned and joined files form the facility-chemical-year data set and outcome, which split into development data and a sealed holdout. The explanatory and predictive paths branch after exploratory analysis, and the holdout was opened once, after the design was locked.

Figure 2*End-to-End Workflow From EPA TRI Files to a Ranked EHS Review List*

Note. Colors mark stage groups, as shown in the legend. Solid arrows show the flow of development data. The dashed box and arrow show the sealed 2023-to-2024 holdout, which was used only for the single final evaluation. CV = cross-validation; EHS = environmental health and safety; RQ = research question; TRI = Toxics Release Inventory.

Workflow Components

Data ingestion: ten EPA text files.

Preprocessing: filtering, consolidation, matching, imputation, capping, and encoding.

Exploratory analysis: [Table 2](#).

Feature engineering: [Table 3](#).

Model development: the RQ2 regression and the baseline and elastic-net RQ4 models.

Model evaluation: a single scoring of the 39,500 holdout records.

Deployment: no production system was built (see Implementation and User Benefit).

Tools and Technologies

Analysis used Python in Google Colab with pandas 2.2.2, NumPy, SciPy, statsmodels 0.14.6, scikit-learn 1.6.1, and matplotlib. The linear models needed no specialized hardware, and deep learning or agentic methods were not used because the questions required interpretable coefficients.

Results

RQ1 to RQ3 used development data, and RQ4 used the sealed holdout for its single evaluation. Very small p values are reported as $p < .001$.

Model Performance

[Table 5](#) shows that the elastic-net model outperformed the baseline in cross-validation (AUC .530 versus .522) and on the holdout (.523 versus .502). The holdout AUC interval, [.515, .532], lies above .50, and the paired difference of .021, 95% CI [.014, .027], excludes zero. Brier scores were nearly identical (.2176 and .2180), so the gain lies in ranking rather than in sharper probabilities.

Table 5*Predictive Model Performance Comparison*

Model	Features Used	Validation Method	CV AUC (Worst Fold)	CV Brier	Holdout AUC [95% CI]	Holdout Brier	Best
Baseline: unpenalized logistic regression	Any reported action; log current release (2 predictors)	Leave-one-transition-out CV; single sealed holdout	.5222 (.5125)	.2173	.5023	.2180	—
Elastic-net logistic regression ($C = 0.01$, L1 ratio = .1)	13 action-year predictors (Table 3)	Leave-one-transition-out CV; single sealed holdout	.5302 (.5234)	.2169	.5231 [.5149, .5316]	.2176	Yes
Difference (elastic net minus baseline)	—	Paired facility bootstrap on holdout	.0080	-.0004	.0208 [.0140, .0268]	-.0004	—

Note. The positive class was non-reduction. Development cross-validation (CV) used 122,017

pound-reported records in three folds grouped by transition. The holdout contained 39,500

pound-reported records, of which 67.95% did not reduce. Confidence intervals for holdout AUC

come from 500 facility-level bootstrap resamples. At the F1-maximizing threshold chosen on

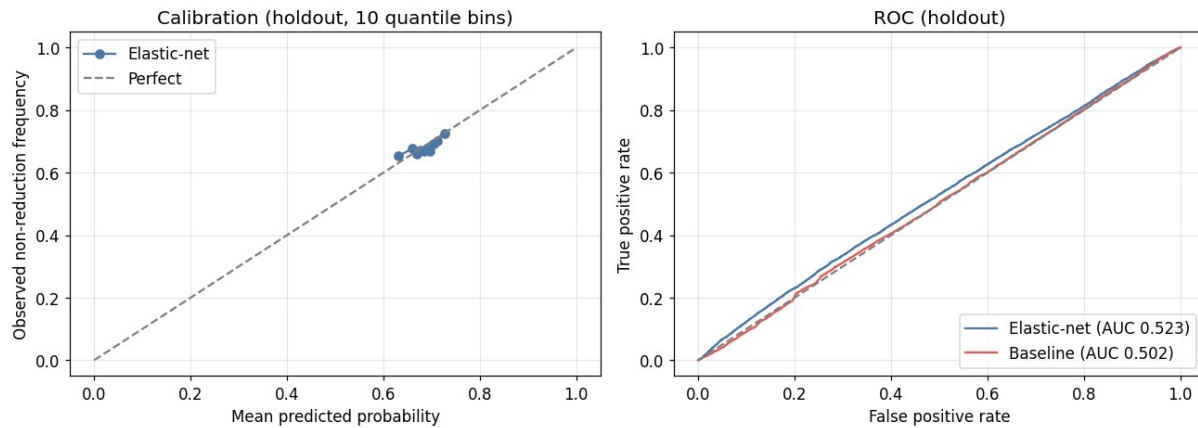
development data (.520), both models flagged nearly every record (elastic net: precision = .679,

recall = .9999, F1 = .809), so accuracy-type metrics at that threshold are uninformative. AUC =

area under the receiver operating characteristic curve. Em dashes indicate not applicable.

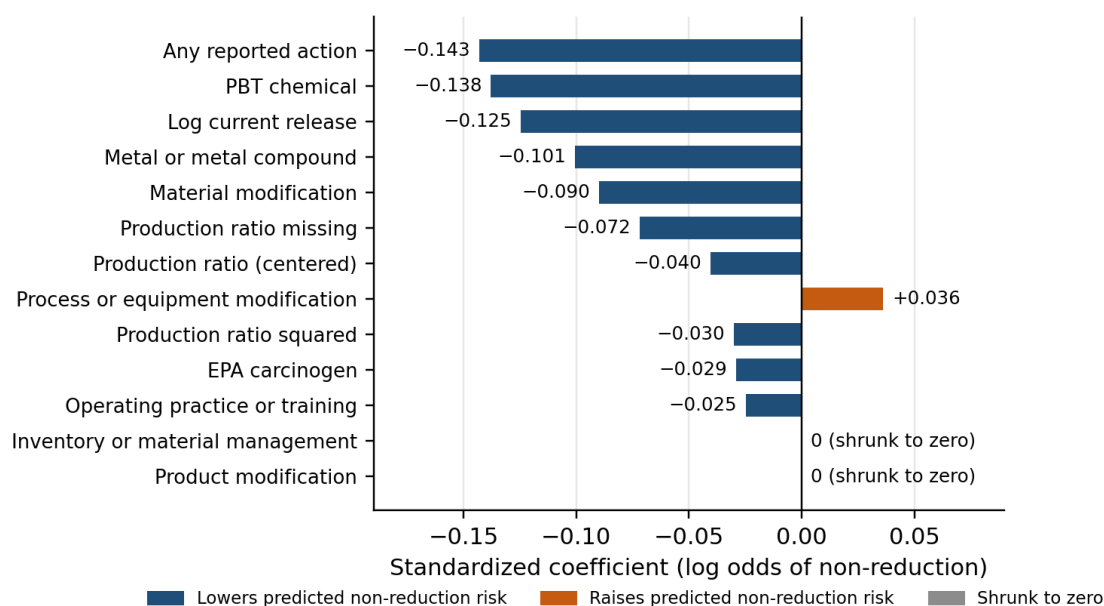
Visual Evidence

In Figure 3, the calibration plot showed no large departures from the diagonal within the narrow range of predicted probabilities (about .63 to .73), although no formal calibration slope was estimated. The receiver operating characteristic (ROC) curve for the elastic net lies only slightly above the baseline and chance lines.

Figure 3*Holdout Calibration and Receiver Operating Characteristic Curves*

Note. Left panel: observed non-reduction frequency against mean predicted probability in 10 quantile bins for the elastic-net model; the dashed line marks perfect calibration. Right panel: ROC curves for the elastic-net model (AUC = .523) and the baseline (AUC = .502) on 39,500 pound-reported 2023-to-2024 holdout records; the dashed line marks chance. AUC = area under the ROC curve.

Figure 4 shows that reporting any action, PBT status, log release, metal status, and material modification lowered predicted risk most. Even the largest coefficient was about -0.14 on the log-odds scale, consistent with modest discrimination (Appendix F, Table F3).

Figure 4*Elastic-Net Standardized Coefficients for Predicting Non-Reduction*

Note. Coefficients are on the log-odds scale of non-reduction. Continuous features were standardized, so coefficients for those features represent a one-standard-deviation change; binary features represent a change from 0 to 1. PBT = persistent, bioaccumulative, and toxic.

Results by Research Question***Research Question 1: Differences by Transition and Subsector***

Hypotheses and Method. H_0 : reduction is independent of transition, and of subsector.

H_a : rates differ. Two chi-square tests with Cramér's V and facility-clustered checks were used.

Result and Decision. Rates rose from 27.5% to 31.1% to 35.9% across transitions, $\chi^2(2) = 663.72, p < .001$, and ranged from 25.9% (paper) to 41.3% (furniture) across subsectors, $\chi^2(14) = 280.04, p < .001$ (Table 6 and Appendix F, Figures F1 and F2). Both null hypotheses were rejected, and both effects held under facility clustering.

Practical Meaning. The effects were statistically significant but small (Cramér's $V = .073$ and $.048$). Time and industry shift the chance of reduction modestly, so they are controls rather than screening levers.

Table 6

RQ1 Reduction Rates by Transition and Subsector

Factor	Reduction Rate, % [95% CI]	χ^2 (df)	p	Cramér's V	Facility-Clustered Check
Action-year transition	2020 to 2021: 27.54 [27.11, 27.97], $n = 41,280$ 2021 to 2022: 31.09 [30.65, 31.54], $n = 41,356$ 2022 to 2023: 35.86 [35.39, 36.32], $n = 40,899$	663.72 (2)	$< .001$	.073	OR = 1.19 and 1.47 versus 2020 to 2021; both $p < .001$
Manufacturing subsector (15 qualifying)	Lowest: 322 Paper, 25.89 [24.82, 27.00] Highest: 337 Furniture, 41.26 [37.92, 44.68] Spread: 15.4 percentage points	280.04 (14)	$< .001$	.048	9 of 14 contrasts with NAICS 325 significant at $p < .05$

Note. Development set, $N = 123,535$ for transitions and 122,429 for the 15 qualifying subsectors.

Rates for all subsectors appear in [Appendix F](#), [Figure F2](#) and [Table F4](#). CI = confidence interval; NAICS = North American Industry Classification System; OR = odds ratio.

Research Question 2: Adjusted Association With Any Reported Action

Hypotheses and Method. H_0 : the adjusted OR for any reported action equals 1. H_a : it differs from 1. A facility-clustered logistic regression was fitted to 122,017 records from 14,836 facilities.

Result and Decision. Reporting any action was associated with higher adjusted odds of meaningful reduction, OR = 1.14, 95% CI [1.06, 1.23], $p < .001$, so H_0 was rejected ([Table 7](#)). The diagnostics did not indicate serious multicollinearity (maximum VIF = 1.98, condition number = 81.0), and the AIC comparison supported the quadratic production-ratio specification (150,795.0 versus 150,817.3).

Practical Meaning. Records reporting an action had about 14% higher *odds*, not 14% higher probability, of reaching the threshold, roughly 3 percentage points at the base rate. This is a modest marker, and because facilities choose whether to act and report, it does not show that the action caused the reduction.

Table 7

RQ2 Adjusted Odds Ratios From the Facility-Clustered Logistic Regression

Term	OR	95% CI	p
Any reported action (exposure)	1.143	[1.058, 1.234]	< .001
Log(1 + current release)	1.042	[1.036, 1.047]	< .001
Production ratio, centered	1.047	[0.977, 1.122]	.197
Production ratio squared	1.085	[1.038, 1.135]	< .001
Production ratio missing	1.108	[0.977, 1.256]	.110
EPA carcinogen	1.020	[0.993, 1.048]	.140
PBT chemical	1.230	[1.167, 1.297]	< .001
Subsector indicators (15 plus Other; reference NAICS 325)	See Table F2	—	—
Transition indicators (reference 2020 to 2021)	See Table F2	—	—

Note. $N = 122,017$ pound-reported records from 14,836 facilities; 38,513 met the reduction threshold. Standard errors were clustered by facility. Diagnostics: maximum variance inflation factor = 1.98; condition number = 81.0; AIC = 150,795.0 for the quadratic form versus 150,817.3 for the linear form. The complete coefficient table appears in [Appendix F, Table F2](#). AIC = Akaike information criterion; CI = confidence interval; NAICS = North American Industry Classification System; OR = odds ratio; PBT = persistent, bioaccumulative, and toxic. Em dashes indicate not applicable.

Research Question 3: Action Types Compared With No Reported Action

Hypotheses and Method. For each action type, H_0 : its reduction rate equals the no-action rate (31.31%). H_a : the rates differ. Five chi-square comparisons used Holm correction and facility-clustered checks.

Result and Decision. Two null hypotheses were rejected (Table 8 and Figure 5).

Material modification reduced at 38.95%, 7.64 points above no action, and operating practice or training at 35.53%, 4.23 points above, both with Holm-adjusted $p < .001$ and both robust to clustering. The other three types did not survive correction.

Practical Meaning. These comparisons are observational and unadjusted, so they show where the clearest signals lie, not what causes reductions. For the other three types, “not statistically demonstrated” does not mean “ineffective,” because their intervals remain compatible with small positive or null differences.

Table 8

RQ3 Reduction Rates by Action Type Compared With No Reported Action

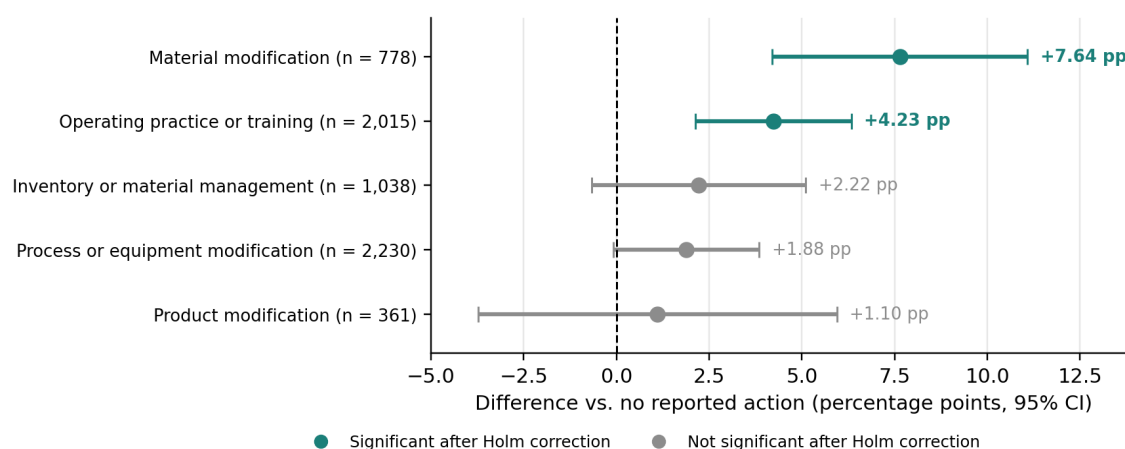
Action Type	<i>n</i>	Rate, % [95% CI]	Difference, pp [95% CI]	RR [95% CI]	Holm <i>p</i>	Clustered OR [95% CI]	H ₀
Material modification	778	38.95 [35.58, 42.42]	7.64 [4.20, 11.08]	1.244 [1.139, 1.359]	< .001	1.400 [1.170, 1.674]	Reject
Operating practice or training	2,015	35.53 [33.47, 37.65]	4.23 [2.12, 6.33]	1.135 [1.070, 1.205]	< .001	1.210 [1.060, 1.379]	Reject
Inventory or material management	1,038	33.53 [30.72, 36.45]	2.22 [−0.66, 5.10]	1.071 [0.983, 1.167]	.27	1.107 [0.930, 1.316]	Retain
Process or equipment modification	2,230	33.18 [31.26, 35.17]	1.88 [−0.09, 3.85]	1.060 [0.999, 1.125]	.18	1.090 [0.968, 1.226]	Retain
Product modification	361	32.41 [27.79, 37.40]	1.10 [−3.73, 5.94]	1.035 [0.892, 1.202]	.69	1.052 [0.812, 1.363]	Retain
No reported action (reference)	117,523	31.31	—	—	—	—	—

Note. Action types are not mutually exclusive, so a record reporting more than one type appears in more than one comparison. The reference group contains only records with no valid action code. The smallest expected cell count was 113, so Pearson chi-square tests were valid.

Clustered odds ratios come from facility-clustered logistic models of each type against the reference group ($p < .001$ for material modification and $p = .005$ for operating practice or training). CI = confidence interval; OR = odds ratio; pp = percentage points; RR = relative risk. Em dashes indicate not applicable.

Figure 5

Difference in Reduction Rate by Action Type Relative to No Reported Action



Note. Points show the percentage-point (pp) difference from the no-action rate of 31.31%, with 95% confidence intervals. Teal marks the two action types that remained significant after Holm correction; gray marks the three that did not.

Research Question 4: Predicting Non-Reduction on the Sealed Holdout

Hypotheses and Method. For RQ4a, H_0 : holdout AUC equals .50. For RQ4b, H_0 : the elastic-net AUC does not exceed the baseline AUC. Each H_a states the reverse. The locked models were scored once on 39,500 holdout records.

Result and Decision. The elastic-net AUC was .523, 95% CI [.515, .532], and it exceeded the baseline by .021, 95% CI [.014, .027], so both null hypotheses were rejected (Table 5).

Practical Meaning. The gain is reliable but modest. An AUC of .523 ranks a non-reducer above a reducer only slightly more often than chance, so the model suits ordering a review queue, not automated classification. A real association (RQ2) can coexist with weak

prediction, because an odds ratio describes an average group difference, while prediction must separate individual records.

Sensitivity Analyses

All six sensitivity checks supported the main conclusions ([Appendix F, Table F1](#)).

Without the 2020-to-2021 transition, the adjusted odds ratio was 1.12, 95% CI [1.03, 1.22], and on complete cases it was 1.14. Clustered versions of RQ1 and RQ3 agreed with the main tests, and dioxins were kept out of the pound-reported models.

Screening and Ranking Results

Because no single cutoff separated the classes well, holdout records were ranked by predicted risk ([Table 9](#) and [Figure 6](#)). Among the top 5%, 73.9% did not reduce, compared with a base rate of about 68%, and the rate faded to 70.7% in the top 30%.

In business terms, a team reviewing about 2,000 of roughly 40,000 records from the top of the list would find about 117 more genuine non-reducers than an unranked review of the same size. This is an approximate operational estimate, not a guaranteed saving.

Table 9

Screening Lift on the Sealed Holdout When Reviewing the Highest-Risk Records

Review Depth	Records Reviewed	Non-Reducers Found	Non-Reduction Rate (Precision)	Lift	Share of All Non-Reducers (Recall)	Additional vs. Unranked Review
Top 5%	1,975	1,459	73.87%	1.087	5.4%	About 117
Top 10%	3,950	2,865	72.53%	1.067	10.7%	About 181
Top 20%	7,900	5,640	71.39%	1.051	21.0%	About 272
Top 30%	11,850	8,376	70.68%	1.040	31.2%	About 324
All records (base rate)	39,500	26,839	67.95%	1.000	100.0%	—

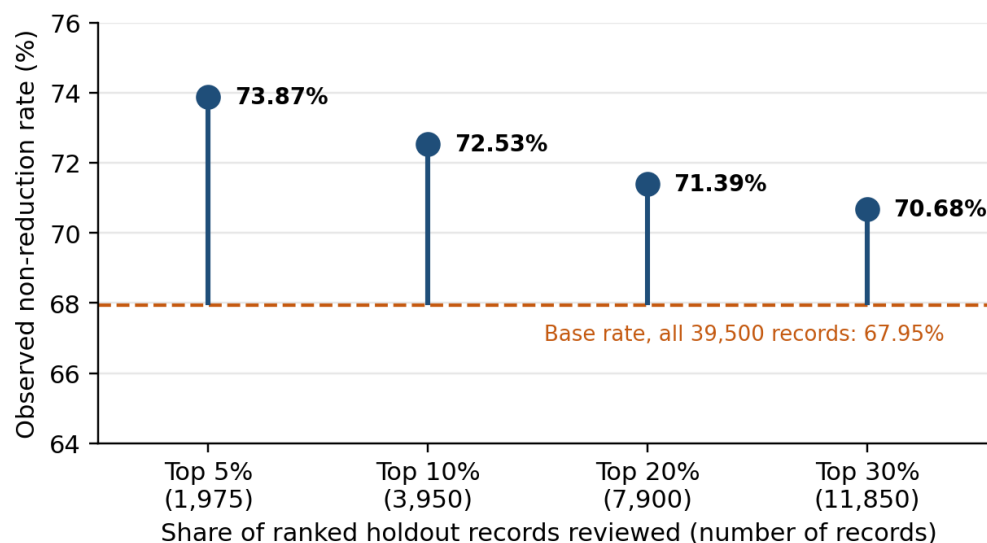
Note. Lift is the non-reduction rate in the reviewed slice divided by the base rate. The last

column compares each ranked slice with the expected yield of an unranked review of the same

size at the base rate; counts were derived from the reported rates and review sizes and are approximate. Em dashes indicate not applicable.

Figure 6

Observed Non-Reduction Rate Among the Highest-Ranked Holdout Records



Note. Records were ranked by the locked elastic-net model's predicted risk of non-reduction.

The dashed line marks the base rate across all 39,500 pound-reported holdout records.

Overall Interpretation

Table 10 summarizes the decisions. Time and industry shift reduction rates, reported actions carry a real but modest adjusted association, two action types show the clearest signals, and the screening model ranks records only slightly better than chance and the baseline.

The main constraint was information content. Actions appear on fewer than 5% of records and record only that a box was checked, not cost, timing, or completion. The heavy penalty selected in tuning and the ample events per parameter point to the same conclusion.

Table 10*Summary of Hypothesis Decisions and Practical Meaning*

RQ	Test	Key Statistic	H ₀ Decision	Size and Practical Meaning
RQ1	Chi-square tests by transition and subsector	$\chi^2(2) = 663.72$; $\chi^2(14) = 280.04$; both $p < .001$	Reject	Small effects ($V = .073$ and $.048$); control for time and industry
RQ2	Facility-clustered logistic regression	$OR = 1.14 [1.06, 1.23]$, $p < .001$	Reject	About 14% higher odds; modest marker, not proof of cause
RQ3	Five Holm-corrected comparisons	Material +7.64 pp; operating practice +4.23 pp; both Holm $p < .001$	Reject for 2 of 5	Clearest signals for further evaluation; other three not demonstrated
RQ4	Elastic net versus chance and baseline on sealed holdout	AUC = .523 [.515, .532] versus .502; difference .021 [.014, .027]	Reject	Reliable but modest gain; suitable for triage, not automation

Note. Brackets show 95% confidence intervals. AUC = area under the receiver operating

characteristic curve; *OR* = odds ratio; pp = percentage points; *V* = Cramér's *V*.

Practical Significance

The results support comparing facilities with peers rather than with raw before-and-after changes, treating a reported action as a modest positive marker, prioritizing the two strongest action types for further evaluation, and ordering fixed review capacity with the ranked list. The model tells a team where to look first, not what to conclude.

Implementation and User Benefit

The results translate into four practical benefits for an EHS team: (a) triage with the model, (b) prioritized evaluation of the strongest action types, (c) weighting by pounds at risk, and (d) benchmarking against peers and trend. All four use existing TRI data, need no new data collection, systems, or authority, and leave every decision with a person. This section describes how the model would be deployed and used and then sets out each benefit with its evidence, its value to users, and its main caution.

Deployment Approach

No production application was built. A realistic deployment is an annual batch process: after each EPA data release, the team runs the existing pipeline, applies the locked feature recipe,

scores each record, and publishes a ranked spreadsheet or dashboard. The model should be re-validated on the latest completed transition before each cycle.

System Integration

The list can join internal systems through TRI facility and chemical identifiers and feed existing review calendars or environmental management systems. A named owner should reassess performance and calibration each reporting cycle and update the model if needed. It should not be used punitively, which could change reporting behavior and weaken the signal.

User Interaction

Reviewers receive one row per record with predicted risk, rank, release quantity, an optional pounds-at-risk score, the subsector benchmark, and the main features behind the score. They examine top-ranked records with site knowledge and record their own decisions.

Benefits to Users

[Table 11](#) summarizes the four benefits. It separates what the data demonstrated from what an organization could do with it and states what the user gains and the main caution for each.

Table 11

Four User Benefits: Evidence, Organizational Use, Value, and Caution

Benefit	What the Data Demonstrated	What an Organization Could Do	What the User Gains	Caution
1. Triage with the model	Top 5% of ranked holdout records: 73.87% non-reduction versus a 67.95% base rate; AUC = .523	Hold review capacity fixed and work through the ranked list from the top each reporting cycle	About 117 more non-reducers found in 1,975 reviews for the same effort	Gain is small and concentrated at the top; not for automated or enforcement decisions
2. Prioritize further evaluation of the strongest action types	Material modification (+7.64 pp) and operating practice or training (+4.23 pp) survived Holm correction and facility clustering	Prioritize these types for further evaluation, small pilots, and knowledge sharing across sites	An evidence-based shortlist for limited prevention and pilot budgets	Comparisons are observational and unadjusted; other types are not proven ineffective
3. Weight by pounds at risk	Every scored record carries a predicted risk and its current release quantity	Order reviews by predicted risk multiplied by release volume	Review effort follows likely environmental impact, not likelihood alone	An operational option; not validated as a performance metric in the study

Benefit	What the Data Demonstrated	What an Organization Could Do	What the User Gains	Caution
4. Benchmark against peers and trend	Rates spanned 15.4 pp across subsectors and rose from 27.54% to 35.86% across transitions	Compare each facility with its subsector peers and recent trend rather than a single universal target	Fairer, like-for-like performance assessment	Effect sizes are small; small subsectors have wide intervals

Note. Counts for Benefit 1 come from [Table 9](#). AUC = area under the receiver operating characteristic curve; pp = percentage points.

Benefit 1: Triage With the Model

Evidence. The locked elastic-net model ranked the 39,500 pound-reported holdout records by predicted risk of non-reduction. Among the top 5% (1,975 records), 73.87% did not reduce, against a base rate of 67.95%, a lift of 1.087. The top 10% reached 72.53%, and the rate moved back toward the base rate with depth ([Table 9](#), [Figure 6](#)).

Value to Users. Review capacity is usually fixed. Working through the ranked list from the top, rather than in an unranked order, would find about 1,459 non-reducers in 1,975 reviews instead of about 1,342, roughly 117 more for the same hours. At the top 10%, the gain is about 181. The benefit is operational efficiency: the same staff time reaches more records that need attention.

Caution. An AUC of .523 is weak discrimination. The gain is small and concentrated at the top of the list, so the ranking should order a queue, not judge individual records or trigger enforcement.

Benefit 2: Prioritize Further Evaluation of the Strongest Action Types

Evidence. Of the five action types, only two remained significant after Holm correction and facility clustering ([Table 8](#)). Records reporting material modification reduced at 38.95%, 7.64 points above the 31.31% no-action rate, clustered $OR = 1.40$, 95% CI [1.17, 1.67]. Records reporting operating practice or training reduced at 35.53%, 4.23 points higher, clustered $OR = 1.21$, 95% CI [1.06, 1.38].

Value to Users. Prevention is not a single intervention. An organization deciding where to direct limited prevention budgets, pilot slots, or cross-site knowledge sharing gains an evidence-based shortlist instead of treating all action types alike. The benefit is strategic: further evaluation starts where the observed signal is clearest.

Caution. These comparisons were observational and unadjusted, so they do not show that either action caused reductions, and the other three types were not shown to be ineffective. Pilots should be evaluated before any wider rollout.

Benefit 3: Weight Reviews by Pounds at Risk

Evidence. Every scored record carried a predicted risk of non-reduction and its current release quantity. Releases were highly skewed, with a mean of 24,737 lb and a median of 217 lb in the development set ([Table 2](#)).

Value to Users. Multiplying predicted risk by current release gives a pounds-at-risk score that turns a statistical ranking into an environmental one. As an illustration, a record with a predicted risk of .65 and 50,000 lb released scores 32,500, whereas a record with a predicted risk of .72 and 100 lb released scores 72. The larger release would be reviewed first even though its predicted risk is lower. The benefit is environmental: review hours follow the likely quantity of pollution involved, not likelihood alone.

Caution. This weighting is an operational option that was not validated as a performance metric in this study, and unweighted pounds do not reflect differences in chemical hazard.

Benefit 4: Benchmark Against Peers and Trend

Evidence. Reduction rates ranged from 25.89% in paper to 41.26% in furniture and related products, a spread of 15.4 points, and rose from 27.54% to 35.86% across transitions

(Appendix F, Table F4). Both differences were statistically significant but small (Cramér's $V = .048$ for subsector and $.073$ for transition).

Value to Users. A single universal target, or a comparison with last year alone, would judge facilities partly by their industry and reporting period. Comparing each facility with its subsector peers and recent trend gives a fairer, like-for-like assessment and reduces the risk of flagging facilities whose rates simply reflect their industry. The benefit is managerial: targets and follow-up rest on credible comparisons.

Caution. Effect sizes were small, and smaller subsectors had wide intervals, such as furniture and related products ($n = 812$, 95% CI [37.92, 44.68]).

Operational, Financial, and Strategic Value

Operational Efficiency. The ranked list lets the same review hours reach about 117 more non-reducers per 1,975 reviews, and peer benchmarks focus follow-up on facilities that lag their own peers.

Financial Impact. No cost, savings, or return on investment was estimated, because TRI does not record the cost of reviews or actions. The financial benefit is indirect: a fixed review budget is directed toward more records that need attention and, with pounds-at-risk weighting, toward larger releases. Quantifying it would require internal cost data.

Strategic Value. Human experts stay in control, because the model tells the team where to look first, not what to conclude. The ranked worklist, the shortlist of action types, and pounds-at-risk scores can also feed green-investment planning, incentive tracking, and sustainability reporting. These are possible uses, not tested results.

Example Use Case

Consider an EHS team with about 40,000 records and capacity for about 2,000 reviews. Using Benefit 1, the team reviews the top 1,975 ranked records and can expect about 1,459 non-reducers rather than about 1,342 from an unranked review. Using Benefit 3, it orders that slice by pounds at risk so the largest releases are examined first. Using Benefit 4, it compares each flagged facility with its subsector trend. Using Benefit 2, where no action was reported, reviewers discuss material modification or operating practice and training as candidates for evaluation. A person makes and records each decision.

Limitations and Further Improvements

The main limitations concern the design, the data, and model performance.

Limitations

Observational Design and Self-Selection. Actions were not assigned, and facilities choose whether to act and to report, so the results are associations and may partly reflect better-managed facilities.

Sparse and Coarse Action Data. Only 4.87% of records reported an action, and TRI does not record cost, timing, completion, capital investment, or process detail. A value of 0 means no code was observed.

Modest Discrimination. A holdout AUC of .523 is only slightly better than chance and is not strong predictive accuracy.

Outcome and Exclusions. REDUCTION_10 cannot separate prevention from production cuts or regression to the mean. Zero-release records and pairs missing the next year were excluded. Attrition was higher among action-reporting records in every transition, ranging from 9.26% to 14.13% versus 7.47% to 9.25% for other records ([Figure A13](#)). Because attrition

differed by action status, the matched action-reporting sample may be selectively biased, and the direction of that bias cannot be determined.

Low Prevalence and Power. Product and material modification were uncommon, and RQ3 was powered for 10-point rather than smaller differences.

Generalizability and Period Effects. Results apply to pound-reported U.S. manufacturing Form R records from 2020 to 2024, and the first transition combined the pandemic with a coding change.

Design Details. Preprocessing constants were not re-estimated within folds, subsector was not tested in RQ4, alternative reduction thresholds such as 5% or 20% were not examined, and calibration was assessed without a formal slope.

Impact of Limitations

These limits mean the odds ratio and action-type differences should guide further evaluation and pilots rather than guarantee results, and the net direction of bias from self-selection and unreported actions is unknown. Sparse action data appear to limit prediction, so the tool is useful for ordering a queue but unreliable for judging a single record or triggering enforcement. The findings may not transfer to other sectors, dioxins, Form A filers, or other periods.

Future Improvements

The main improvements concern data rather than algorithms. Priority additions are more precise production data, the cost, timing, and completion status of each action, plant and process characteristics, capital and sustainability investment, and multi-year action histories. With richer data, the model can be re-evaluated with fold-level preprocessing, subsector indicators, alternative reduction thresholds, and formal calibration and decision-curve analysis.

Future Scope

New reporting years can be added to re-validate the model and monitor drift, a facility-level model could support site planning, and the approach could extend to other sectors. Causal questions would require data on action timing and assignment that TRI does not provide.

Conclusion

This study asked how far recent TRI action information can explain and predict meaningful next-year reductions in U.S. manufacturing. The answer is: to a limited but real degree.

Reduction rates differed modestly by year and subsector (RQ1). After adjustment, reporting any action was associated with about 14% higher odds of reduction, OR = 1.14, 95% CI [1.06, 1.23] (RQ2), and material modification and operating practice or training showed the clearest signals (RQ3). The elastic-net model beat chance and a simple baseline on the sealed holdout (AUC = .523 versus .502), but only modestly (RQ4).

Two lessons stand out. A real association can coexist with weak prediction, so explanation and prediction must be judged separately. The main constraint on performance appears to be the limited information content of TRI action fields. In practice, the results give EHS teams four benefits: triage of fixed review capacity, a shortlist of action types for further evaluation, pounds-at-risk weighting, and fair peer and trend benchmarks. Within these limits, the study offers a transparent decision-support tool for deciding where to look first, not a decision maker.

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Appendix A

Extended Exploratory Data Analysis

This appendix expands on [Table 2](#) in the main text. The figures were produced by the exploratory data analysis notebook in the project repository using development data only, and every proportion carries a Wilson score interval (Agresti & Coull, 1998). Charts from that notebook keep their original panel labels.

Rare Exposure and a Coding-Scheme Change

Only 6,012 of 123,535 development records (4.87%) reported any prevention action, while 31.48% met the reduction threshold ([Figure A1](#)). Rare exposure, not class imbalance, was therefore the main statistical constraint. The EPA also replaced legacy W codes with S codes after 2020 ([Figure A2](#)). Process or equipment modification rose from 1.32% to 2.13% and material modification from 0.41% to 0.87% across that boundary, while the other three types stayed flat ([Figure A3](#)). Because the jump coincides with the scheme change, part of it may reflect coding rather than behavior, so the key association was re-estimated without the 2020-to-2021 transition.

Industry Composition and a Confounded Raw Gap

Subsectors that reported the most actions were not those that reduced most often. Nonmetallic mineral products reported actions on 1.63% of records but had the second-highest reduction rate (35.04%), while computer and electronic products reported actions on 12.02% of records and reduced at 30.24% ([Figure A4](#)). Records reporting any action reduced at 34.91% ($n = 6,012$), compared with 31.31% for records reporting none ($n = 117,523$), a gap of 3.61 percentage points, 95% CI [2.37, 4.84] ([Figure A5](#)). However, seven of 19 covariates differed by more than a standardized mean difference of 0.10, and the two largest differences (0.244 and

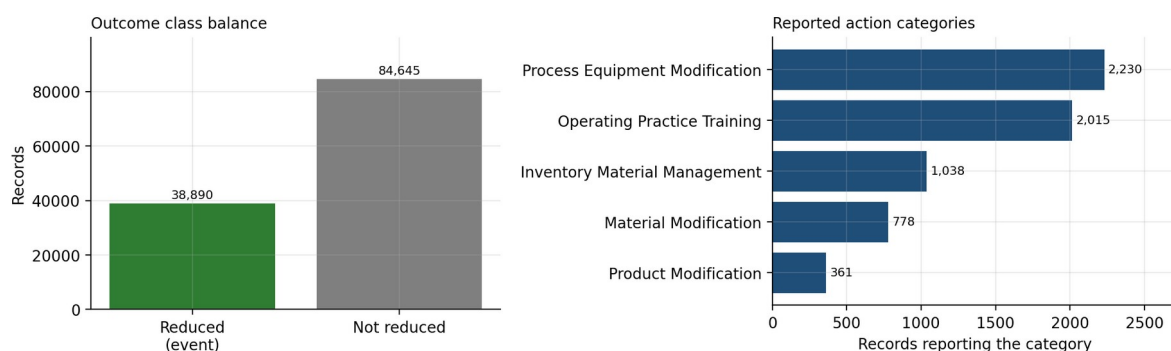
0.232) reflected industry composition (Figure A6). The gap was positive in every displayed stratum, but several stratum intervals included zero (Figure A7). The raw gap could not be interpreted until industry, release level, and year were adjusted, which is the RQ2 question.

Skewed Releases and the Production Ratio

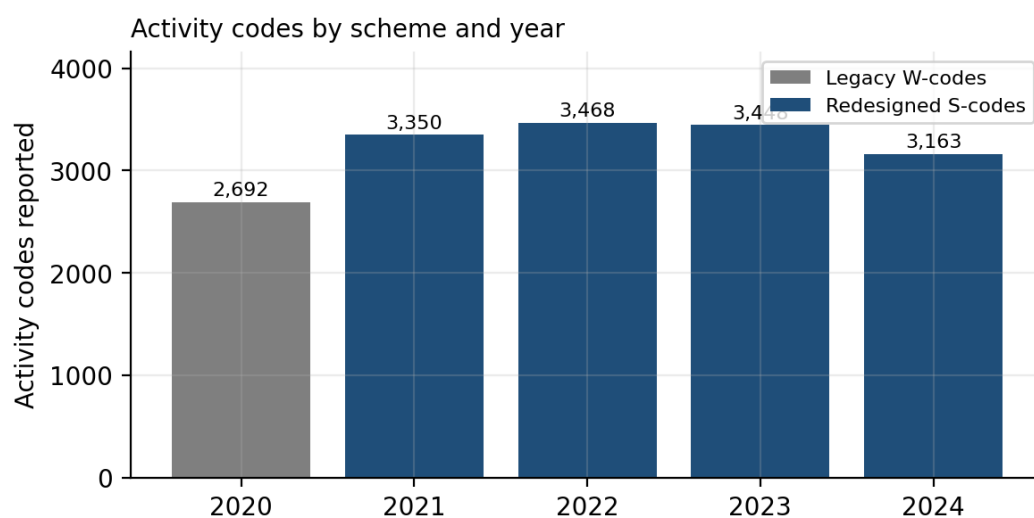
Across 122,017 pound-reported records, the mean current release was 24,737 lb against a median of only 217 lb, with a maximum of 19,750,398 lb and skewness of 34.6 (Figure A8). The extremes were genuine large emitters and were kept. Reduction rates did not rise steadily with release size (Figure A9), which is consistent with regression to the mean (Barnett et al., 2005), and the empirical logit rose broadly with log release (Figure A10). Release therefore entered every model on the log scale as a required control. The production ratio was centered near 1.00, was missing for 1.62% of records, and contained extreme but valid values up to 3,149,500 (Figure A11). Its U-shaped relationship with the outcome is shown in Figure 1 of the main text.

Repeated Facilities, Dioxins, and Attrition

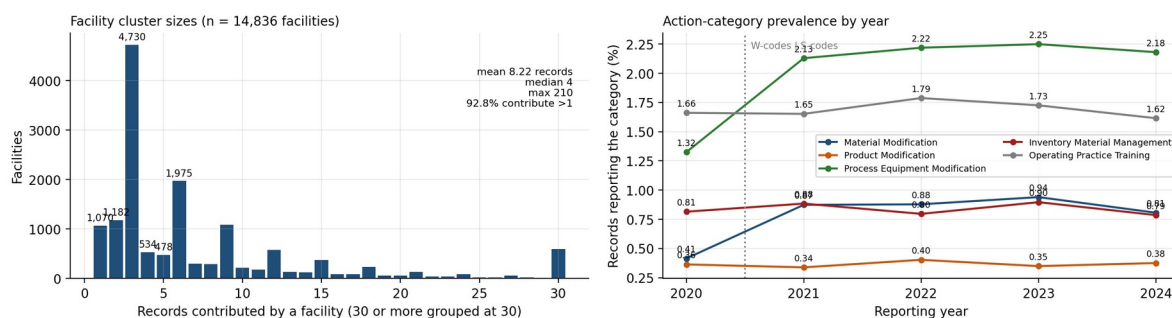
The 122,017 pound-frame records came from 14,836 facilities, an average of 8.22 records per facility (median 4, maximum 210), and 92.8% came from facilities contributing more than one record (Figure A3). Treating these records as independent would understate standard errors (Cameron & Miller, 2015), so facility clustering was a requirement. Dioxins reduced at 24.84%, compared with 31.56% for other chemicals, and were analyzed separately because they are reported in grams (Figure A12). Between 7.6% and 9.5% of facility-chemical pairs did not reappear the following year, with higher attrition among action-reporting records (Figure A13). Of 15 data-quality checks, eight passed without corrective action, five identified issues that were controlled analytically, and two remained limitations of the source data (Figure A14).

Figure A1*Outcome Class Balance and Reported Action Categories*

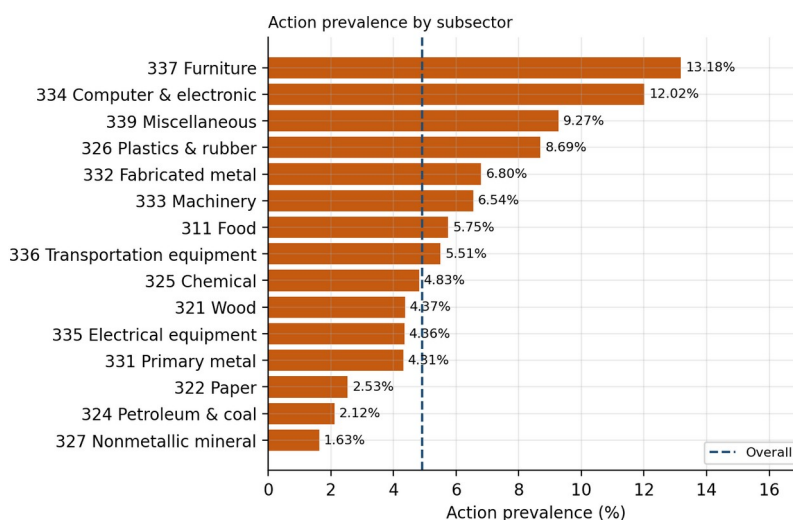
Note. Left: of 123,535 development records, 38,890 (31.48%) met the reduction threshold. Right: records reporting each action category; categories are not mutually exclusive. Only 6,012 records (4.87%) reported any action.

Figure A2*Source-Reduction Activity Codes by Coding Scheme and Reporting Year*

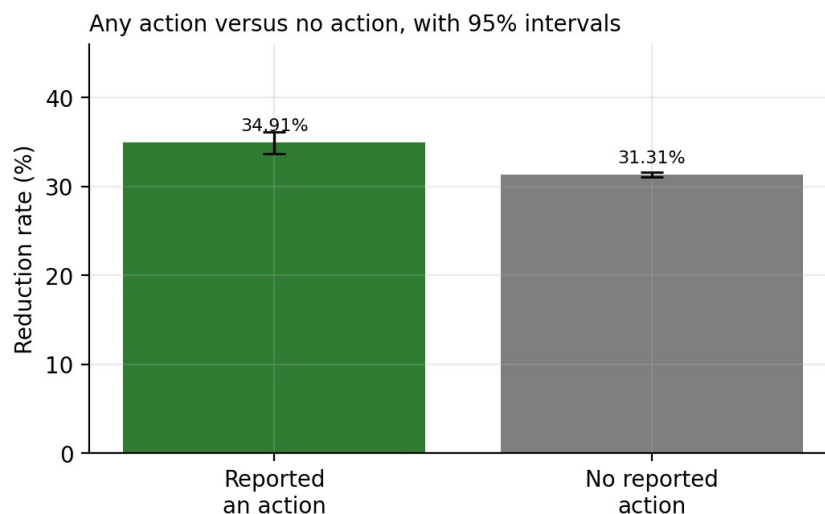
Note. Legacy W codes appear in 2020 and are replaced by redesigned S codes from 2021 onward. All 70 distinct codes were mapped to the five action types, with no unmapped codes. Reporting year 2024 File 2A was used only for this descriptive check.

Figure A3*Facility Cluster Sizes and Action-Category Prevalence by Reporting Year*

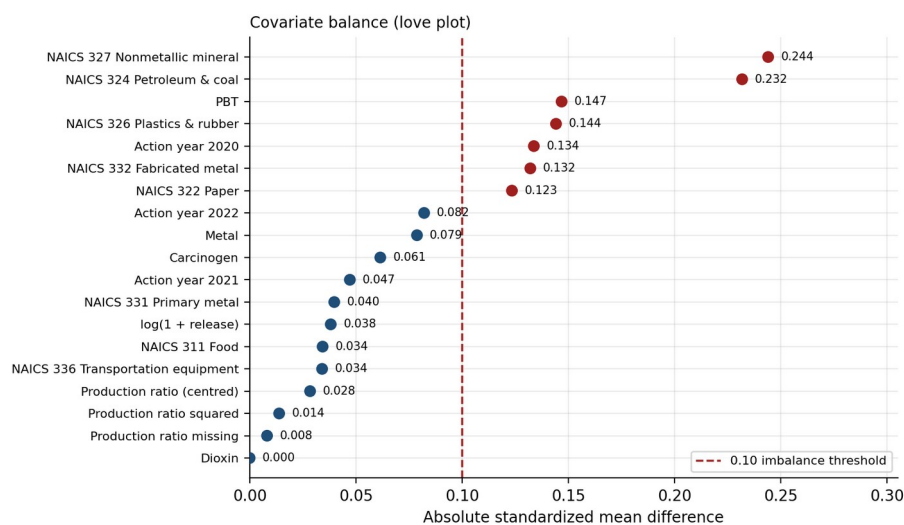
Note. Left: records contributed per facility in the pound-reported frame (mean 8.22, median 4, maximum 210); 92.8% of records come from facilities contributing more than one. Right: process or equipment modification rises from 1.32% in 2020 to 2.13% in 2021 and material modification from 0.41% to 0.87%, coinciding with the coding-scheme change.

Figure A4*Action Prevalence by Manufacturing Subsector*

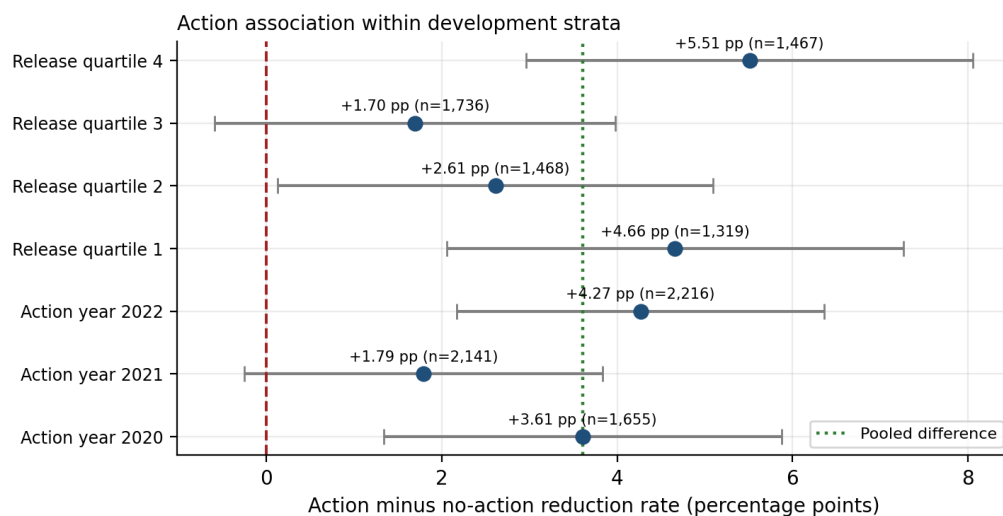
Note. The dashed line marks overall action prevalence. Subsectors reporting the most actions are not those with the highest reduction rates (compare [Appendix F, Figure F2](#)).

Figure A5*Unadjusted Reduction Rate for Any Reported Action Versus No Reported Action*

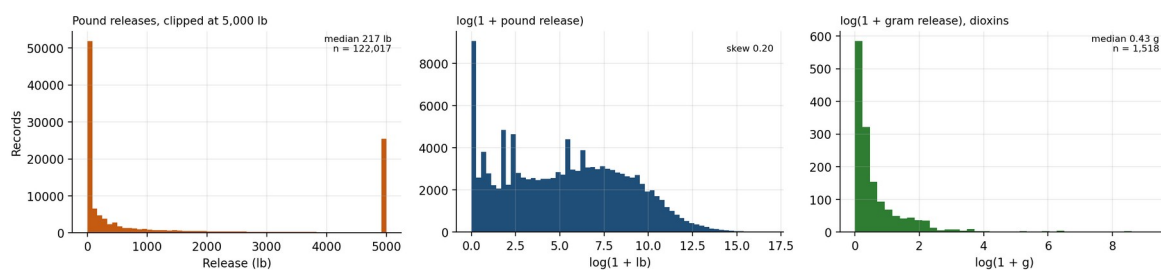
Note. Error bars show 95% Wilson confidence intervals. The unadjusted gap is 3.61 percentage points, 95% CI [2.37, 4.84]. This comparison is not adjusted for industry, release level, or year.

Figure A6*Covariate Balance Between Action and No-Action Groups*

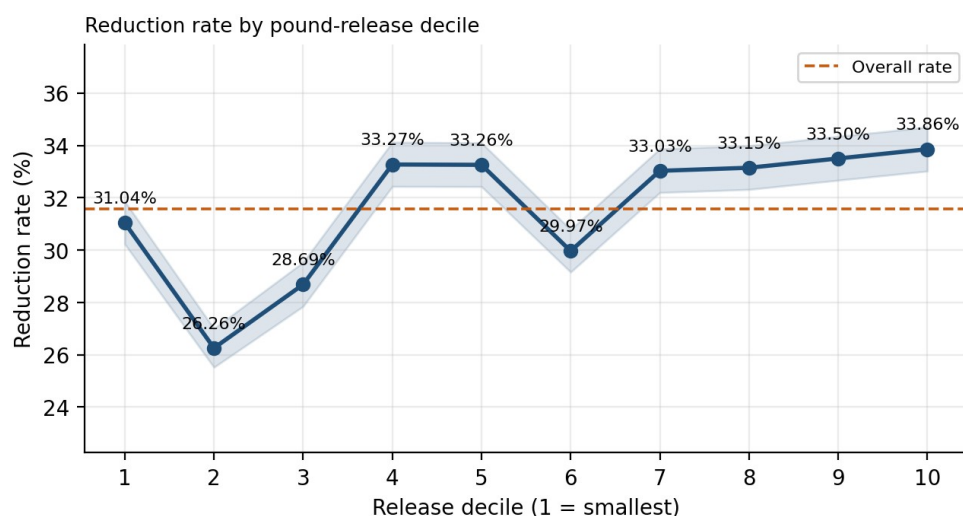
Note. Seven of 19 covariates exceed an absolute standardized mean difference of 0.10 (dashed line). The two largest differences, 0.244 and 0.232, reflect industry composition.

Figure A7*Unadjusted Action Association Within Development Strata*

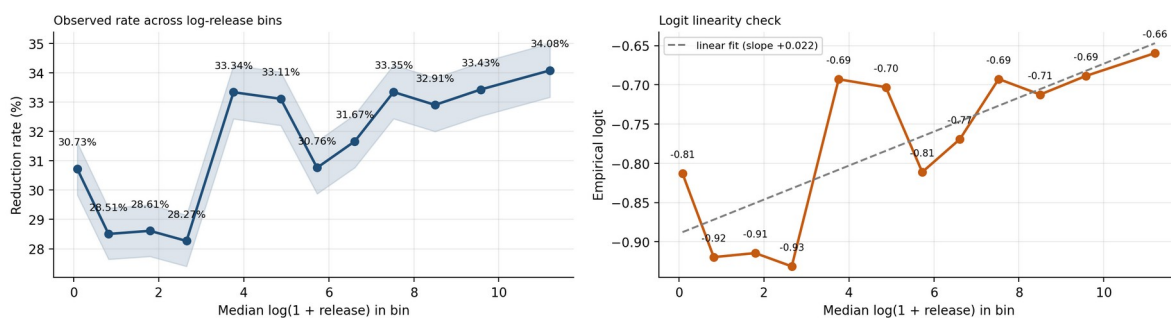
Note. Point estimates are positive in every displayed stratum, but several confidence intervals include zero. These are exploratory consistency checks, not separate evidence of significance.

Figure A8*Distribution of Current-Year Releases*

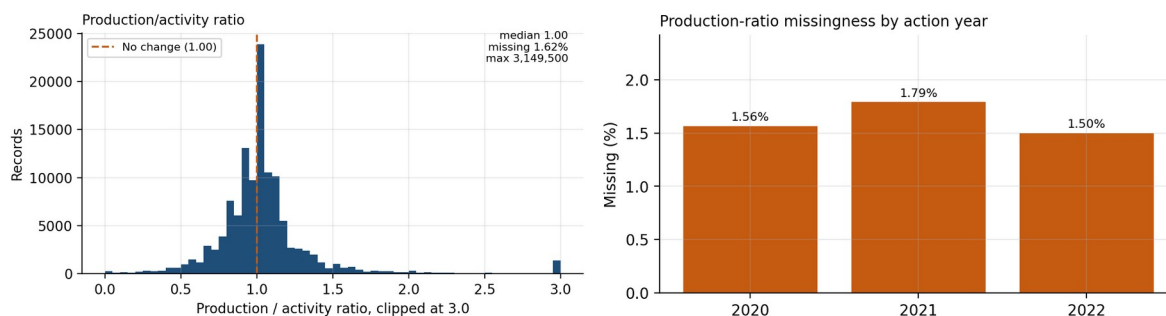
Note. Pound-reported releases are shown clipped at 5,000 lb and on a log scale; gram-reported dioxins are shown separately. Skewness of the raw pound releases is 34.6, with a median of 217 lb and a mean of 24,737 lb.

Figure A9*Reduction Rate by Current-Release Decile*

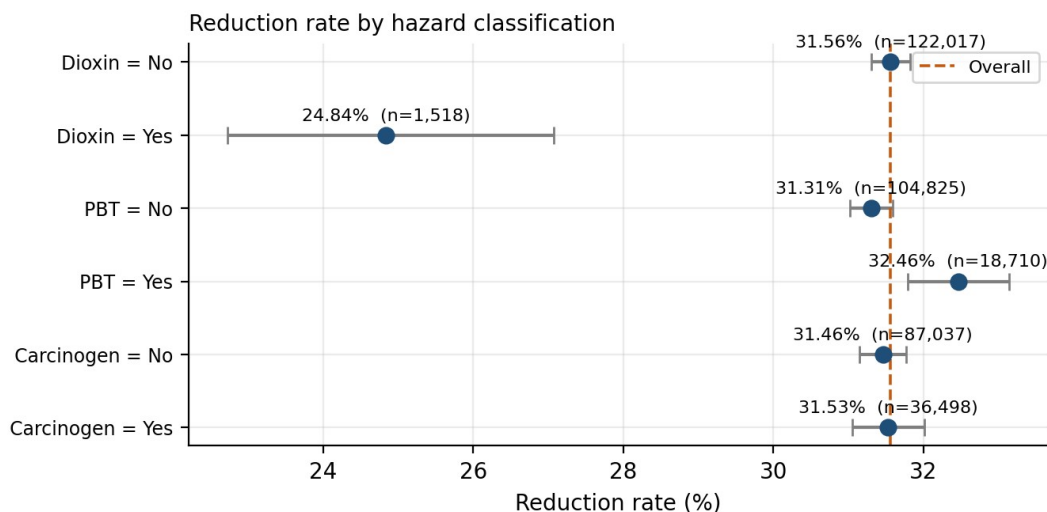
Note. Shaded bands are 95% Wilson confidence intervals. Rates are generally higher in upper deciles, but the pattern is not monotonic, which is consistent with starting level and regression to the mean shaping the room for a percentage reduction.

Figure A10*Observed Reduction Rate Across Log-Release Bins With a Logit Linearity Check*

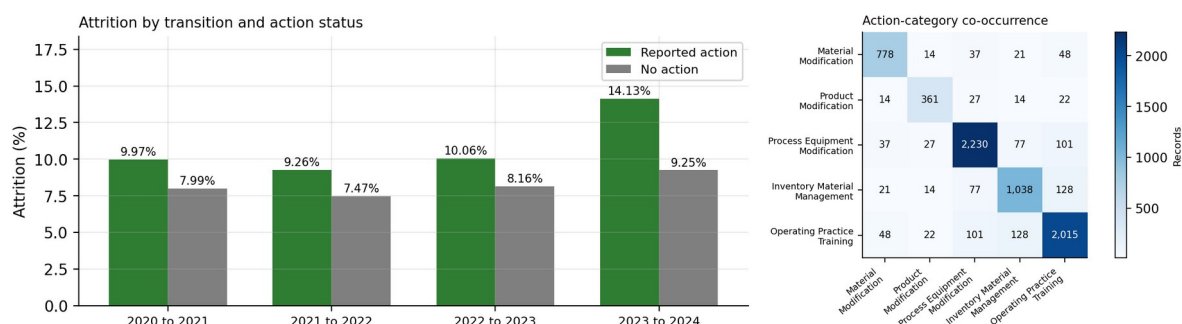
Note. The empirical logit generally increases with log current release, supporting the log transformation. Some departures from linearity remain; a spline comparison is listed as a future check.

Figure A11*Distribution of the Production Ratio and Its Missingness by Action Year*

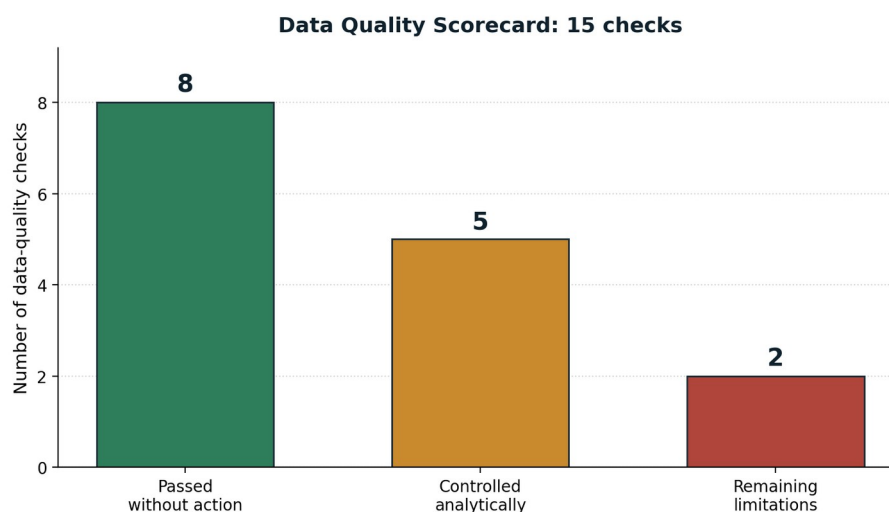
Note. Left: the ratio is centered on 1.00 and is clipped at 3.0 for display; the maximum observed value was 3,149,500. Values were capped at the development 99th percentile (3.15) before centering and squaring. Right: missingness ranges from 1.50% to 1.79% across action years; the missingness mechanism cannot be confirmed from observed data.

Figure A12*Reduction Rate by Chemical Hazard Classification*

Note. Error bars show 95% Wilson confidence intervals. Carcinogen status shows a difference of 0.07 percentage points. Dioxins reduce about 6.7 points less often than other chemicals and were analyzed separately.

Figure A13*Cross-Year Attrition and Action-Category Co-Occurrence*

Note. Left: overall attrition ranges from about 7.6% to 9.5% across transitions; within the groups shown, attrition ranges from 7.47% to 14.13% and is higher for action-reporting records in every transition. Right: diagonal cells show records per action category, and off-diagonal cells show co-reporting; records reporting an action average 1.07 distinct categories.

Figure A14*Data-Quality Scorecard Outcomes*

Note. Of 15 data-quality checks, eight passed without corrective action, five identified issues that were controlled analytically, and two (cross-year attrition and TRI coverage limits) remain limitations of the source data.

Appendix B

Source File Provenance, Parsing Rule, and Repository Structure

[Table B1](#) lists the size and SHA-256 checksum of each source file. All ten checksums are distinct, confirming that no file was duplicated during download, and any reader can verify their own copies against these values. [Table B2](#) records the duplicate-consolidation counts by reporting year. File 1A data lines carry 283 fields against a 282-field header because each line ends with a trailing delimiter, so fields were read by position with that offset applied.

Table B1

Source File Sizes and SHA-256 Checksums

Reporting Year	File Type	Size (MB)	SHA-256 (First 16 Characters)
2020	1A	62.6	b94f8cb1286d7420
2020	2A	44.5	5ddaa1624fbba793
2021	1A	62.2	73daa53931a58696
2021	2A	44.4	805fbd9c932f51aef
2022	1A	63.6	4cf809c08d11cd8c
2022	2A	45.4	16f382966622b5e8
2023	1A	62.6	0c60fa25c4ac4348
2023	2A	44.7	cc3017cd0ca0b310
2024	1A	61.6	f463ec07ff320968
2024	2A	44.0	dbedc8bafc05338a

Note. Checksums are truncated for readability. Full 64-character values are printed by the exploratory notebook and stored in the repository. The ten files total 535.6 MB.

Table B2

Duplicate Consolidation Counts by Reporting Year

Year	Before	After	Removed	Groups With Duplicates	Revision Superseded	Originals Summed
2020	57,004	56,462	542	321	33	288
2021	56,750	56,219	531	353	39	314
2022	56,725	56,155	570	367	13	354
2023	55,882	55,349	533	333	26	307
2024	54,595	54,115	480	304	16	288

Note. Unit conflicts and chemical-identifier conflicts were zero in every year. Summing dominates supersession, which is why the consolidation rule in [Appendix C](#) sums original filings rather than keeping a single filing.

Repository Structure

The public GitHub repository is organized as follows:

- README.md: problem statement, data source, record counts, and reproduction instructions
- requirements.txt and VALIDATION_CHECKLIST.md: software requirements and validation notes
- data/: download instructions and a 200-row sample of File 1A and File 2A for each reporting year
- notebooks/: the early data exploration notebook, the exploratory data analysis notebook that produced the [Appendix A](#) figures, and the final analysis notebook that produced the RQ1 to RQ4 results
- src/data_prep.py: the data-preparation pipeline, including parsing, consolidation, matching, and export
- reports/: the project synopsis and earlier project documents

Appendix C

Duplicate-Filing Consolidation Rule

A facility sometimes files more than one Form R for the same chemical in the same reporting year. The original study plan took the release quantity from the filing with the highest document control number. Implementation showed that this rule was wrong for the most common case, so the rule was revised before any analysis. This appendix records the final rule and the evidence for it.

Duplicate filings arise for two different reasons. In the first, a facility submits a revised form that replaces an earlier one. The revision genuinely supersedes the original, so keeping one filing is correct. In the second and much more common case, a facility submits several original forms that each carry part of the year's total for that chemical. Keeping a single filing would discard real quantity and understate the release. Across the five reporting years, 1,551 duplicate groups were of the second kind and only 127 were of the first.

The final rule is therefore conditional. If one or more filings in a facility-chemical-year group carry a value in either revision-code field, the revision-coded filing with the highest document control number supplies the release quantity and unit, and the other filings are treated as superseded. Source-reduction indicators are taken from that filing, and the production or activity ratio is taken from it when available. If no revision-coded filing exists, the release quantities of the original filings are summed after unit and chemical-identifier consistency checks. The sum uses a minimum-count setting of one, so a group in which every value is missing produces a missing total rather than a false zero. If more than one unit appeared in a group, the release would be set to missing. No such conflict occurred in any year.

Two notes accompany the rule. First, adding the second revision-code field and the minimum-count safeguard changed no results. Revision counts were identical with one field or two, and no group consisted entirely of missing values. Both were kept as defensive measures. Second, an earlier planning remark that duplicate original forms represent separate process units was withdrawn, because the TRI fields available in this study cannot establish that. The forms are summed because summing preserves the reported total, which is defensible whatever the reason for the duplicates. The same rule was applied when the 2024 outcome for the sealed holdout was reconstructed, which yielded 944 superseded and 52,670 summed facility-chemical records among 2024 pound-reported manufacturing filings.

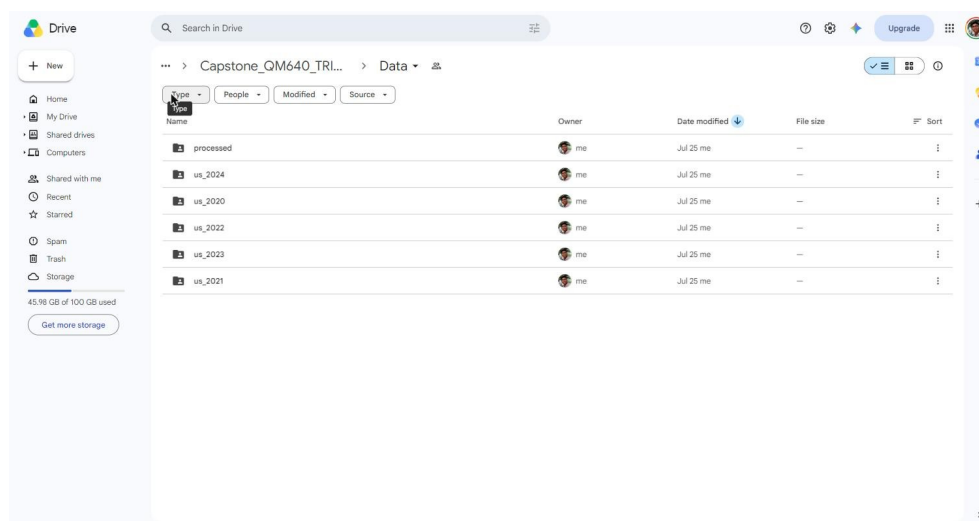
Appendix D

Data Availability Evidence

Figures D1 to D4 document the shared folder structure and data set contents described in the Code and Data Availability statement and the Data Source section. Access is set to anyone with the link, with viewer permission.

Figure D1

Raw Data Folder Structure on Google Drive



Note. One folder per TRI reporting year, 2020 through 2024, each holding File 1A and File 2A, plus the processed subfolder. TRI = Toxics Release Inventory.

Figure D2*Processed Data Folder Contents*

Name	Owner	Date modified	File size	IP Sort
activity_code_crosswalk.csv	me	Jul 26 me	7 KB	:
audit_attrition.csv	me	Jul 26 me	359 bytes	:
audit_consolidation.csv	me	Jul 26 me	308 bytes	:
audit_field_widths.csv	me	Jul 26 me	285 bytes	:
audit_join.csv	me	Jul 26 me	514 bytes	:
audit_malformed_rows.csv	me	Jul 26 me	320 bytes	:
data_dictionary.csv	me	Jul 26 me	1 KB	:
development.csv	me	Jul 26 me	16.1 MB	:
preprocessing_parameters.json	me	Jul 26 me	3 KB	:
quality_scorecard.csv	me	Jul 26 me	1 KB	:
reserved_2023_2024_features_only.csv	me	Jul 26 me	4.3 MB	:

Note. Eleven files. The holdout file (4.3 MB) is smaller than the development data set (16.1 MB) because it carries no outcome columns.

Figure D3*Preview of the Processed Development Dataset*

development.csv																	Open with	Share
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	
1	facility_id_new	chemical_id_new	cas_number_new	chemical_name_new	subsector_new	year	volume_new	production_vol_new	vol_ratio_new	%_carbogen_new	%_oil_new	%_ethanol_new	%_methanol_new	reported_any_action	material_modified	product_modified	process_modified	
2	150117LQ0P0BQ1N123	NI20		Dibenzotriazoles	320	2020, 2021	20	1	0	0	0	0	0		0	0	0	
3	150117N0T230H	100435	10042-0	Ethylene	320	2020, 2021	39540	0.9	0	1	0	0	0		0	0	0	
4	902200P0N0B0U0L0	7440473	744047-3	Chromium	331	2020, 2021	7730	1	0	0	0	0	0		0	0	0	
5	902200P0N0B0U0L0	7430095	743009-5	Manganese	331	2020, 2021	240171	1	0	0	0	0	0		0	0	0	
6	902200P0N0B0U0L0	7440020	744002-0	Nickel	331	2020, 2021	108106	1	0	0	0	0	0		0	0	0	
7	973900K0N0R0T0E0E0	100414	10041-4	Ethylbenzene	320	2020, 2021	899	0.22	0	0	0	0	0		1	1	0	
8	973900K0N0R0T0E0E0	100414	10041-0	Toluene	320	2020, 2021	4239	0.15	0	0	0	0	0		0	0	0	
9	973900K0N0R0T0E0E0	130007	1300-207	Xylene (methyl ester)	320	2020, 2021	644	0.21	0	0	0	0	0		0	1	0	
10	974100L0M0K0R0E0	7440058	7440-50-0	Copper	331	2020, 2021	116	1.06	0	0	0	0	0		1	1	0	
11	974100L0M0K0R0E0	NA20		Lead compounds	331	2020, 2021	33	1.04	0	0	1	0	0		1	1	0	
12	904600L0K0I0C0A0	NA20	NA20	Carbon (graphite)	345	2020, 2021	5	0.64	0	0	0	0	0		0	0	0	
13	904600L0K0I0C0A0	50000	50-00-0	Formaldehyde	325	2020, 2021	100990	1.02	0	0	0	0	0		0	0	0	
14	904600L0K0I0C0A0	7897372	7897-37-2	Nitric acid	5	125	0	0	0	0	0	0	0		0	0	0	
15	904600L0K0I0C0A0	109892	109-82-0	Phenol	330	2020, 2021	11149	0.79	0	0	0	0	0		0	0	0	
16	904600L0K0I0C0A0	7440473	7440-47-3	Chromium	331	2020, 2021	7730	1	0	0	0	0	0		0	0	0	
17	904600L0K0I0C0A0	7440058	7440-09-0	Copper	333	2020, 2021	0.93	0.76	0	0	0	0	0		0	1	0	
18	904600L0K0I0C0A0	7440020	7440-02-0	Nickel	330	2020, 2021	0.93	0.76	0	1	0	0	0		0	0	0	
19	904600L0K0I0C0A0	191242	191-24-2	Benzotriazole	324	2020, 2021	0.51	1	0	0	0	0	0		0	0	0	
20	904600L0K0I0C0A0	NA20	NA20	Polycyclic aromatic	334	2020, 2021	2000	1	1	0	0	0	0		0	0	0	
21	904600L0K0I0C0A0	7439921	7439-92-1	Lead	330	2020, 2021	0.3	0.98	0	1	1	0	0		0	0	0	
22	904600L0K0I0C0A0	79010	79-01-0	Trichloroethylene	332	2020, 2021	29020	0.98	0	0	0	0	0		0	0	0	
23	904600L0K0I0C0A0	778304	7783-08-54	Hydrogen sulfide	324	2020, 2021	899.13	0	0	0	0	0	0		0	0	0	
24	904600L0K0I0C0A0	778304	7783-08-54	Hydrogen sulfide	324	2020, 2021	844.17	0	0	0	0	0	0		0	0	0	
25	904600L0K0I0C0A0	3507395	3507-39-5	Dibenzotriazoles	325	2020, 2021	10	1.31	0	0	0	0	0		0	0	0	
26	904600L0K0I0C0A0	901115	0961-11-5	Tetrachloroethylene	325	2020, 2021	10	0.91	0	0	0	0	0		0	0	0	
27	904600L0K0I0C0A0	2045331	2045-33-1	Permethrin	325	2020, 2021	10	1.4	0	0	0	0	0		0	0	0	
28	904600L0K0I0C0A0	91203	91-20-3	Formaldehyde	325	2020, 2021	10	1.66	0	0	0	0	0		0	0	0	
29	904600L0K0I0C0A0	7439921	7439-92-1	Lead	330	2020, 2021	0.35	0.93	0	1	0	0	0		0	0	0	
30	904600L0K0I0C0A0	NA20	NA20	Lead compounds	331	2020, 2021	589	1	0	0	0	0	0		0	0	0	
31	904600L0K0I0C0A0	7439976	7439-97-6	Mercury	331	2020, 2021	10	1	0	0	0	0	0		0	0	0	
32	904600L0K0I0C0A0	94186	94-18-6	Formic acid	310	2020, 2021	3907	0.71	0	0	0	0	0		0	0	0	
33	904600L0K0I0C0A0	NA20	NA20	Dibenzotriazoles	332	2020, 2021	9	2.11	0	0	0	0	0		0	0	0	
34	904600L0K0I0C0A0	NA20	NA20	Lead compounds	325	2020, 2021	30	0.71	0	0	0	0	0		1	1	0	
35	904600L0K0I0C0A0	100420	100-42-0	Ethylene	320	2020, 2021	127790	0	0	1	0	0	0		0	0	0	
36	904600L0K0I0C0A0	7440473	7440-47-3	Chromium	330	2020, 2021	0.98	1.06	0	0	0	0	0		1	1	0	
37	904600L0K0I0C0A0	7440058	7440-09-0	Copper	330	2020, 2021	0.93	0.76	0	0	0	0	0		0	0	0	

Figure D4*Activity Code Crosswalk With EPA Descriptions*

Code	Category	EPA description	2020	2021	2022	2023	2024	Total	Status
601	material_modification	SUBSTITUTED A PL	0	29	22	20	10	81	mapped
602	material_modification	SUBSTITUTED ANE	0	36	41	31	28	136	mapped
603	material_modification	SUBSTITUTED RUA	0	177	168	185	100	630	mapped
604	material_modification	SUBSTITUTED WHA	0	63	61	62	61	247	mapped
605	material_modification	MODIFIED CONTIN	0	66	69	41	34	210	mapped
606	material_modification	OTHER MATERIAL I	0	102	117	143	132	594	mapped
607	product_modification	REFORMULATED O	0	96	111	90	90	388	mapped
608	product_modification	ALTERED DIMENSIO	0	21	21	30	27	100	mapped
609	product_modification	MODIFIED PRODUCT	0	10	7	0	0	17	mapped
610	product_modification	OTHER PRODUCT S	0	61	77	66	78	282	mapped
611	process_equipments	OPTIMIZED PROCE	0	555	577	540	632	2404	mapped
612	process_equipments	INSTITUTED PROC	0	36	67	46	43	192	mapped
613	process_equipments	IMPLEMENTED ME	0	146	180	115	122	630	mapped
614	process_equipments	MODIFIED OR UPGR	0	204	189	233	185	811	mapped
615	process_equipments	OTHER PROCESS S	0	239	286	214	189	907	mapped
616	inventory_materials	INSTITUTED BETTE	0	63	71	76	61	271	mapped
617	inventory_materials	CHANGED SIZE OR	0	17	8	11	7	43	mapped
618	inventory_materials	IMPROVED CONTAI	0	77	45	52	71	245	mapped
619	inventory_materials	IMPROVED MONITO	0	72	123	81	67	343	mapped
620	inventory_materials	OTHER APPROVISE	0	242	286	282	204	1014	mapped
621	operating_practices	IMPROVED SCHED	0	368	405	405	337	1515	mapped
622	operating_practices	CHANGED PRODUCT	0	63	109	103	93	368	mapped
623	operating_practices	INTRODUCED RUA	0	26	24	21	23	94	mapped
624	operating_practices	OTHER APPROVISE	0	453	462	452	442	1809	mapped
625	operating_practices	IMPROVED MAINT	453	2	1	1	0	457	mapped
626	operating_practices	CHANGED PRODUCT	114	0	0	0	0	114	mapped
627	operating_practices	INTRODUCED MA	50	0	0	0	0	50	mapped
628	operating_practices	OTHER CHANGES	304	0	0	0	0	304	mapped
629	inventory_materials	INSTITUTED PROC	79	0	0	0	0	79	mapped
630	inventory_materials	BEGAN TO TEST O	12	0	0	0	0	12	mapped
631	inventory_materials	FLAMMABLE SHED	6	0	0	0	0	6	mapped
632	inventory_materials	INSTITUTED BETTE	2	0	0	0	0	2	mapped
633	inventory_materials	INSTITUTED CLEAF	2	0	0	0	0	2	mapped
634	inventory_materials	OTHER CHANGES	40	0	0	0	0	40	mapped
635	inventory_materials	IMPROVED FORM	23	0	0	0	0	23	mapped
636	inventory_materials	IMPROVED PROCESS	49	0	0	0	0	49	mapped

Note. Every observed code is mapped to one of the five action types, with counts by year.

Legacy W codes carry counts only in 2020, and S codes carry counts only from 2021 onward.

EPA = U.S. Environmental Protection Agency.

Appendix E

Data and Method Supporting Tables

These tables support the Materials and Method section. [Table E1](#) defines the analysis variables, [Table E2](#) records the cleaning decisions, [Table E3](#) reconciles record counts, [Table E4](#) gives the sample-size calculations, and [Table E5](#) lists the evaluation metrics.

Table E1

Data Dictionary for the Analysis Variables

Variable	Definition	Type	Range or Values	Missing-Value Handling	Role
facility_id	TRI facility identifier (TRIFD)	Text	14,851 distinct facilities	Never missing	Identifier
chemical_id	TRI chemical identifier	Text	507 distinct chemicals	Never missing	Identifier
cas_number	Chemical Abstracts Service number	Text	As reported	Retained as reported	Identifier
chemical_name	EPA chemical name	Text	As reported	Retained as reported	Identifier
release_now	Current-year total on-site releases	Numeric	Greater than 0; maximum 19,750,398 lb	Zero and missing values excluded at eligibility	Predictor
release_next	Next-year total on-site releases	Numeric	0 or greater	Unmatched records excluded, never set to zero	Outcome input
reduced (REDUCTION_10)	1 if release_next is at most $0.90 \times \text{release_now}$	Binary	0 or 1	Defined only when release_now is greater than 0	Outcome
percent_change	Percentage change in release	Numeric	Unbounded	Defined only when release_now is greater than 0	Outcome (descriptive)
production_ratio_now	Production or activity ratio, current year	Numeric	Median 1.00	Development median imputed; capped at 3.15; complete-case sensitivity completed	Predictor
ratio_missing	1 when the production ratio was absent	Binary	0 or 1	Derived, never missing	Predictor
subsector_now	First three digits of the primary NAICS code	Categorical	15 qualifying subsectors plus Other Manufacturing	Record dropped if outside 31 to 33	Predictor
span	Action-year to outcome-year transition	Categorical	Four transitions	Derived, never missing	Predictor (RQ2)
reported_any_action_now	1 if any valid source-reduction code is present	Binary	0 or 1	Blank codes read as 0 (not reported)	Predictor
Five action-type indicators	Material, product, process or equipment, inventory or material management, operating practice or training	Binary	0 or 1; not mutually exclusive	Blank codes read as 0	Predictor
is_carcinogen_now	EPA carcinogen classification	Binary	0 or 1	As supplied by EPA	Predictor
is_pbt_now	Persistent, bioaccumulative, and toxic classification	Binary	0 or 1	As supplied by EPA	Predictor

Variable	Definition	Type	Range or Values	Missing-Value Handling	Role
is_dioxin_now	Dioxin or dioxin-like compound	Binary	0 or 1	As supplied by EPA	Descriptive
is_metal_now	Metal or metal compound	Binary	0 or 1	As supplied by EPA	Predictor (RQ4)
unit_of_measure_now	Pounds or grams	Categorical	Pounds or grams	Case normalized; record excluded if units differ across years	Audit

Note. A value of 0 on any action indicator means that no valid code was observed in the four File 2A activity-code fields. It does not prove that the facility took no environmental action. EPA = U.S. Environmental Protection Agency; NAICS = North American Industry Classification System; TRI = Toxics Release Inventory.

Table E2

Data Cleaning Log

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Non-Form R filings	All	Form type field	35,155 rows removed	Form A carries no release quantity
Out-of-scope industries	Primary NAICS code	First two digits of NAICS	54,065 rows removed	Study scope is manufacturing (31 to 33)
Duplicate facility-chemical forms	All	Grouping on facility, chemical, and year	2,176 rows consolidated	Separate forms would double-count
Revised filings	Revision code fields 1 and 2	Either revision field non-blank	127 groups: revision kept, others superseded	A revision replaces the original
Multiple original forms	Total on-site releases	No revision code in group	1,551 groups summed with minimum count of 1	Preserves the total; avoids false zeros
Unit label case	Unit of measure	Value inspection	Labels upper-cased before frame construction	Prevents 864 pound records from being misclassified
Unit conflicts within a group	Unit of measure	More than one unit in a group	None found; release would be set to missing	Pounds and grams must never be added
Unit change between years	Unit of measure	Comparison across the matched pair	Record excluded from eligibility	A unit change makes a percentage change meaningless
Zero or missing base release	release_now	Value check	41,946 records excluded (with unit changes)	A percentage change needs a nonzero base
Unmatched facility-chemical pairs	Join keys	Left join to following year	18,738 records excluded	No outcome; not treated as a reduction
Missing production ratio	production_ratio_now	Numeric coercion	Development median (1.00) imputed; ratio_missing flag; complete-case check run	Keeps records while preserving missingness information
Extreme production ratio	production_ratio_now	Percentiles and maximum	Capped at development 99th percentile (3.15) before centering and squaring	Prevents isolated extreme values from dominating
Extreme release skew	release_now	Distribution inspection	Retained; log(1 + release) used in models	Real large emitters, not data errors
Activity-code scheme change	Activity code fields	Cross-tabulation of code by year	70 codes mapped to five types; zero unmapped asserted	One behavior would otherwise carry two labels
Outcome leakage risk	Holdout outcome columns	Schema assertion	Outcome columns stripped from reserved file; labels rebuilt after lock	Protects the out-of-time evaluation

Note. NAICS = North American Industry Classification System.

Table E3

Record Flow From Raw Files to Analysis Sets

Stage	Records	Removed	Reason
Raw File 1A rows, action years 2020 to 2023	315,581	—	Reporting year 2024 read only to build outcomes
Form R only	280,426	35,155	Form A carries no release quantity
Manufacturing only (NAICS 31 to 33)	226,361	54,065	Study scope
Joined one-to-one to File 2A	226,361	0	Every Form R record matched one File 2A record
After duplicate consolidation	224,185	2,176	Multiple forms per facility-chemical-year combined (Appendix C)
Matched to the following year	205,447	18,738	The facility-chemical pair must recur to have an outcome
Eligible: positive current release, same unit	163,501	41,946	A percentage change needs a nonzero base and a common unit
Development set (2020 to 2023 transitions)	123,535	—	Outcome available; used for RQ1 to RQ3 and model development
Pound-reported modeling frame	122,017	—	RQ2 and RQ4 development; 14,836 facilities
Gram-reported dioxin records	1,518	—	Analyzed separately
Sealed holdout (2023 to 2024 transition)	39,966	—	Outcome withheld until the model was locked
Pound-reported RQ4 modeling holdout	39,500	—	Single RQ4 evaluation
Gram-reported holdout records	466	—	Outside the release-based model

Note. The development and holdout sets sum exactly to the eligible total (123,535 + 39,966 = 163,501). The development set contains 14,851 distinct facilities. NAICS = North American Industry Classification System. Em dashes indicate not applicable.

Table E4

Minimum Sample-Size Requirements Compared With Available Records

RQ	Planning Method	Assumptions	Minimum	Available	Margin
RQ1	95% CI for a proportion: $n = Z^2 p(1 - p)/e^2$	$p = .50$, $Z = 1.96$, $e = .03$	1,068	123,535	116×
RQ2	Power approximation for a binary association	OR = 1.30, $\alpha = .05$, power = .80, exposure 4.87%, no-action rate .3138	10,756	122,017	11.3×
RQ3	Two-proportion comparison per action type	10-point difference, $\alpha = .01$ for five comparisons, power = .80	About 290 each	361 to 2,230	1.2× to 7.7×
RQ4	Events per parameter (Peduzzi et al., 1996)	Planning: 20 parameters, 20 events each, event rate .3155	1,269	122,017 (38,513 events)	96×

Note. The locked RQ4 model had 13 parameters and 38,513 events in the smaller outcome class, about 2,963 events per parameter, which also satisfies the more detailed criteria of Riley et al. (2019) and van Smeden et al. (2019) in spirit. Clustering by facility reduces the effective sample size, so the facility-clustered intervals in the Results are the direct evidence of precision. CI = confidence interval; OR = odds ratio.

Table E5

Evaluation Metrics and Formulas

Metric	Formula	Interpretation	Used For
Reduction rate	$p = x / n$	Share of records meeting the threshold	RQ1, RQ3
Wilson score interval	$[p + z^2/2n \pm z \sqrt{(p(1-p)/n + z^2/4n^2)}] / (1 + z^2/n)$	Interval for a proportion; reliable at extreme values	RQ1, RQ3
Pearson chi-square	$\chi^2 = \Sigma (O - E)^2 / E$	Tests whether rates differ across groups	RQ1, RQ3
Cramér's V	$V = \sqrt{\chi^2 / [n \times \min(r - 1, c - 1)]}$	Effect size for a chi-square test	RQ1
Rate difference	$d = p_1 - p_0$	Percentage-point gap versus no action	RQ3
Relative risk and Katz interval	$RR = p_1 / p_0$; $SE(\ln RR) = \sqrt{(1/a - 1/n_1 + 1/c - 1/n_0)}$	Ratio of reduction rates with a log-scale interval	RQ3
Holm-adjusted p	Step-down: compare ordered $p_{(i)}$ with $\alpha / (m - i + 1)$	Controls family-wise error across five tests	RQ3
Odds ratio	$OR = \exp(\beta)$	Adjusted association	RQ2
Clustered standard error	Sandwich estimator grouped by facility	Accounts for repeated facility records	RQ1 to RQ3
Variance inflation factor	$VIF = 1 / (1 - R_j^2)$	Detects collinearity	RQ2
Akaike information criterion	$AIC = 2k - 2 \ln(L)$	Compares functional forms; lower is better	RQ2
AUC	Area under the true-positive versus false-positive rate curve	Ranking ability; .50 is chance	RQ4
Brier score	$B = (1/n) \Sigma (\hat{y}_i - y_i)^2$	Mean squared error of probabilities	RQ4
Precision	$TP / (TP + FP)$	Share of flagged records that truly do not reduce	RQ4
Recall	$TP / (TP + FN)$	Share of non-reducers that are flagged	RQ4
F1 score	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$	Balances precision and recall	RQ4
Lift at depth q	Non-reduction rate in top q / base rate	Gain from ranked review	RQ4
Calibration curve	Observed versus mean predicted rate in 10 quantile bins	Agreement of probabilities with outcomes	RQ4
Facility bootstrap interval	Percentile interval over 500 resamples of whole facilities	Uncertainty that respects clustering	RQ4

Note. In RQ4 the positive class is non-reduction. TP = true positives; FP = false positives; FN = false negatives. AUC = area under the receiver operating characteristic curve.

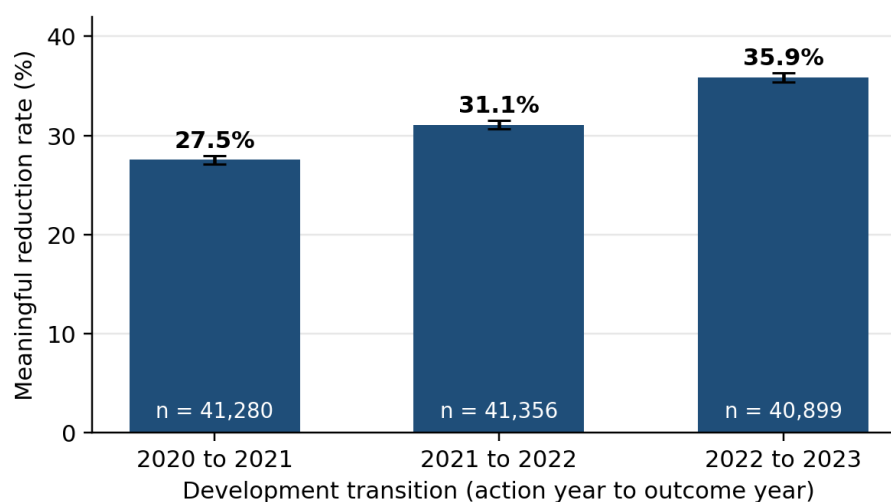
Appendix F

Supplementary Results

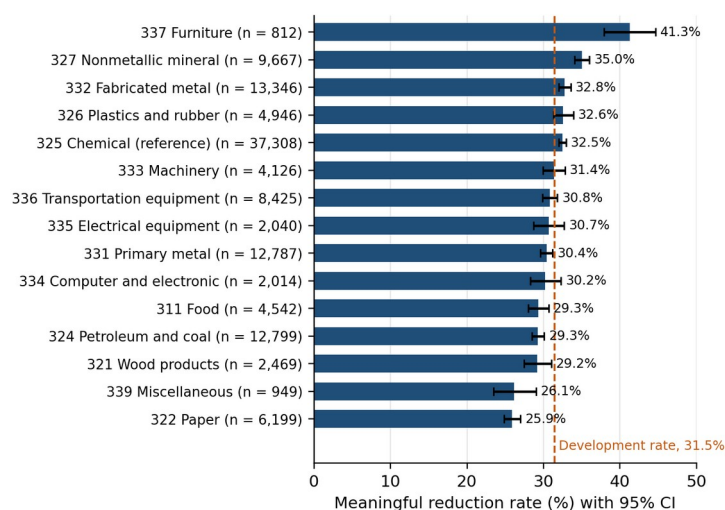
This appendix holds supporting results referenced in the Results section: the RQ1 charts by transition and subsector (Figures F1 and F2), the six sensitivity analyses (Table F1), the complete RQ2 model (Table F2), the locked elastic-net coefficients (Table F3), and reduction rates for every qualifying subsector (Table F4).

Figure F1

Meaningful Reduction Rate by Development Transition



Note. Error bars show 95% Wilson confidence intervals. The 2023-to-2024 transition is excluded because it was the sealed holdout.

Figure F2*Meaningful Reduction Rate by Manufacturing Subsector*

Note. Subsectors are three-digit NAICS codes with at least 500 development records. Error bars show 95% Wilson confidence intervals, and the dashed line marks the overall development rate. NAICS = North American Industry Classification System.

Table F1*Sensitivity Analyses and Their Effect on the Conclusions*

Analysis	Concern Addressed	N	Key Result	Conclusion
RQ2 without the 2020-to-2021 transition	Pandemic disruption and coding-scheme change	81,246	OR = 1.124 [1.033, 1.223], $p = .006$	Association stable
RQ2 complete cases for production ratio	Median imputation of 1.62% missing ratios	120,045	OR = 1.139 [1.054, 1.231], $p = .001$	Not an imputation artifact
RQ2 functional form	Linear versus quadratic production ratio	122,017	AIC difference = 22.35 in favor of the quadratic form	Quadratic term retained
RQ1 facility-clustered models	Repeated records per facility	123,535	Transition ORs 1.19 and 1.47, $p < .001$; 9 of 14 subsector contrasts significant	RQ1 conclusion holds
RQ3 facility-clustered models	Repeated records per facility	Per comparison	Material modification OR = 1.400; operating practice or training OR = 1.210; other three not significant	RQ3 conclusion holds
Gram-reported dioxins analyzed separately	Grams and pounds cannot be combined	1,518	24.84% [22.73, 27.07] versus 31.56% [31.30, 31.82] in the pound frame	Findings limited to pound-reported chemicals

Note. Brackets show 95% confidence intervals. AIC = Akaike information criterion; OR = odds ratio.

Table F2*Complete RQ2 Adjusted Logistic Regression With Facility-Clustered Standard Errors*

Term	OR	95% CI	p
Intercept (baseline odds)	0.301	[0.283, 0.320]	< .001
Any reported action	1.143	[1.058, 1.234]	< .001
Log(1 + current release)	1.042	[1.036, 1.047]	< .001
Production ratio, centered	1.047	[0.977, 1.122]	.197
Production ratio squared	1.085	[1.038, 1.135]	< .001
Production ratio missing	1.108	[0.977, 1.256]	.110
EPA carcinogen	1.020	[0.993, 1.048]	.140
PBT chemical	1.230	[1.167, 1.297]	< .001
311 Food	0.821	[0.753, 0.896]	< .001
321 Wood products	0.804	[0.706, 0.915]	< .001
322 Paper	0.658	[0.586, 0.739]	< .001
324 Petroleum and coal products	0.857	[0.791, 0.928]	< .001
326 Plastics and rubber products	0.988	[0.914, 1.068]	.763
327 Nonmetallic mineral products	1.140	[1.040, 1.249]	.005
331 Primary metal	0.941	[0.876, 1.011]	.097
332 Fabricated metal products	1.077	[1.011, 1.148]	.021
333 Machinery	1.068	[0.966, 1.180]	.199
334 Computer and electronic products	0.922	[0.813, 1.046]	.206
335 Electrical equipment	0.988	[0.863, 1.132]	.866
336 Transportation equipment	0.951	[0.879, 1.028]	.205
337 Furniture and related products	1.394	[1.157, 1.679]	< .001
339 Miscellaneous manufacturing	0.757	[0.647, 0.887]	< .001
Other manufacturing	1.179	[1.020, 1.363]	.026
Transition 2021 to 2022	1.179	[1.123, 1.238]	< .001
Transition 2022 to 2023	1.465	[1.399, 1.535]	< .001

Note. $N = 122,017$ pound-reported development records from 14,836 facilities. Subsector

reference: 325 Chemical manufacturing. Transition reference: 2020 to 2021. CI = confidence

interval; EPA = U.S. Environmental Protection Agency; OR = odds ratio; PBT = persistent,

bioaccumulative, and toxic.

Table F3*Locked Elastic-Net Coefficients for Predicting Non-Reduction*

Predictor	Standardized Coefficient	Odds Multiplier
Any reported action	−0.143	0.867
PBT chemical	−0.138	0.871
Log(1 + current release)	−0.125	0.883

Predictor	Standardized Coefficient	Odds Multiplier
Metal or metal compound	−0.101	0.904
Material modification	−0.090	0.914
Production ratio missing	−0.072	0.930
Production ratio, centered	−0.040	0.961
Process or equipment modification	0.036	1.037
Production ratio squared	−0.030	0.970
EPA carcinogen	−0.029	0.971
Operating practice or training	−0.025	0.975
Product modification	0.000	1.000
Inventory or material management	0.000	1.000

Note. Positive class = non-reduction, so values below 0 (odds multipliers below 1) lower

predicted risk. Continuous predictors were standardized. Selected hyperparameters: $C = 0.01$ and $L1 \text{ ratio} = .1$. EPA = U.S. Environmental Protection Agency; PBT = persistent, bioaccumulative, and toxic.

Table F4

Reduction Rate by Manufacturing Subsector

NAICS	Subsector	<i>n</i>	Rate (%)	95% CI
322	Paper	6,199	25.89	[24.82, 27.00]
339	Miscellaneous manufacturing	949	26.13	[23.44, 29.02]
321	Wood products	2,469	29.20	[27.44, 31.03]
324	Petroleum and coal products	12,799	29.28	[28.50, 30.08]
311	Food	4,542	29.33	[28.02, 30.67]
334	Computer and electronic products	2,014	30.24	[28.27, 32.28]
331	Primary metal	12,787	30.39	[29.60, 31.19]
335	Electrical equipment	2,040	30.69	[28.72, 32.72]
336	Transportation equipment	8,425	30.81	[29.84, 31.81]
333	Machinery	4,126	31.36	[29.96, 32.79]
325	Chemical	37,308	32.46	[31.99, 32.94]
326	Plastics and rubber products	4,946	32.57	[31.28, 33.89]
332	Fabricated metal products	13,346	32.77	[31.98, 33.57]
327	Nonmetallic mineral products	9,667	35.04	[34.09, 35.99]
337	Furniture and related products	812	41.26	[37.92, 44.68]

Note. Development records in the 15 subsectors with at least 500 records ($N = 122,429$). $\chi^2(14) =$

280.04, $p < .001$, Cramér's $V = .048$. CI = 95% Wilson confidence interval; NAICS = North

American Industry Classification System.

Appendix G

Supplementary Literature Relevance Matrix

Table G1 covers the supporting sources not shown in Table 1. Author names link to the References.

Table G1

Literature Relevance Matrix for Supporting Sources

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	How It Supports the RQs or Project Decisions
Agresti & Coull (1998)	Binomial interval estimation	Statistical comparison	Coverage comparison	Score-type intervals such as the Wilson interval outperform the simple normal approximation.	All reduction-rate intervals use the Wilson method
Lin et al. (2013)	Large-sample inference	Information systems research	Methodological commentary	With very large samples, p values approach zero for trivial effects.	Report Cramér's V and intervals with every test
Barnett et al. (2005)	Regression to the mean	Epidemiological measurement	Methodological review	Extreme baseline values tend to move toward the mean; adjusting for baseline helps.	Force current release into every model
Aiken & West (1991)	Regression with nonlinear terms	Regression methodology	Centering	Centering before squaring reduces artificial collinearity.	Centered quadratic production-ratio term
Akaike (1974); Burnham & Anderson (2002)	Model selection	Statistical theory	Information criteria	AIC differences above about 10 leave essentially no support for the weaker model.	AIC test of quadratic versus linear form (RQ2)
Peduzzi et al. (1996)	Logistic regression sample size	Simulation study	Events-per-variable simulation	Too few events per variable produce biased coefficients.	Events-per-parameter floor for RQ4
Riley et al. (2019); van Smeden et al. (2019)	Prediction model sample size	Derivation and simulation	Sample-size criteria	Required size depends on outcome rate, predictor strength, and shrinkage.	RQ4 sample-size check and regularization
Van Calster et al. (2019)	Model calibration	Methodological review	Review	Poorly calibrated models can mislead decisions even with acceptable discrimination.	Calibration check before screening use
EPA (2025a); EPA (2025b)	Pollution prevention policy; national TRI analysis	Federal program information	Program documentation	Describe the source-reduction hierarchy and national TRI trends.	Policy context for the business problem
EPA (2025c); EPA (2025d)	TRI Basic Plus files and guides	Federal data documentation	Program documentation	Define every field, activity code, and file layout used.	Authoritative source for parsing, codes, and units

Note. AIC = Akaike information criterion; EPA = U.S. Environmental Protection Agency; TRI =

Toxics Release Inventory.